Name

Mathematics Stage 5 Paths

Circle Geometry

Book 1

Version: 261008

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Contents

Syllabus Content 2
Learning Intention 48

Syllabus Content

MA5-CIR-P-01 applies deductive reasoning to prove circle theorems and solve related problems

Prove and apply angle and chord properties of circles

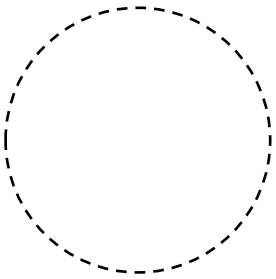
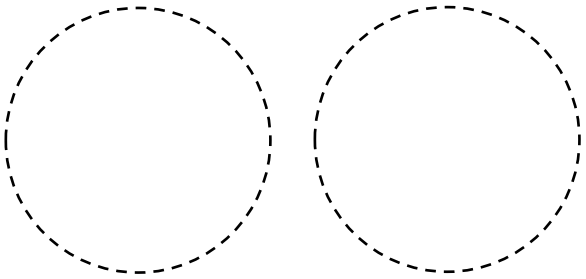
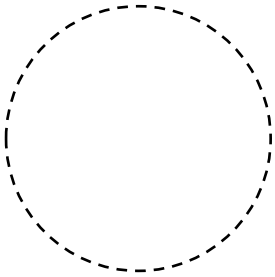
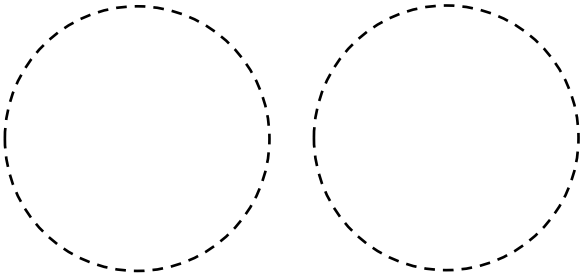
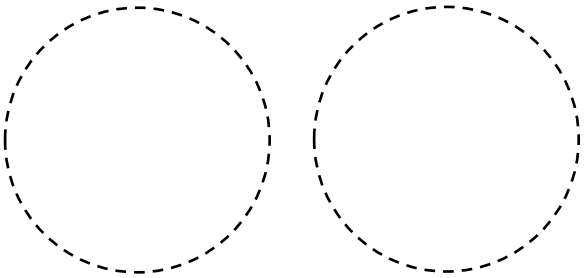
- Apply terminology associated with angles in circles
- Identify the arc on which an angle at the centre or circumference stands
- Demonstrate that at any point on a circle there is a unique tangent to the circle, and that this tangent is perpendicular to the radius at the point of contact
- Prove chord properties of circles
- Prove angle properties of circles
- Apply chord and angle properties of circles to find unknown angles and lengths in diagrams

Prove and apply tangent and secant properties of circles

- Prove tangent and secant properties of circles
- Apply tangent and secant properties of circles to find unknown angles and lengths in diagrams

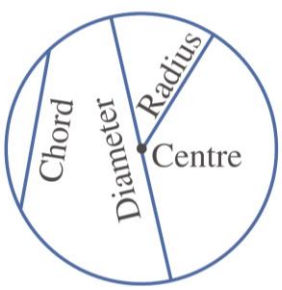
Features of Circles

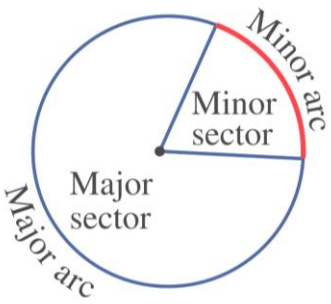
Identify Features of circles

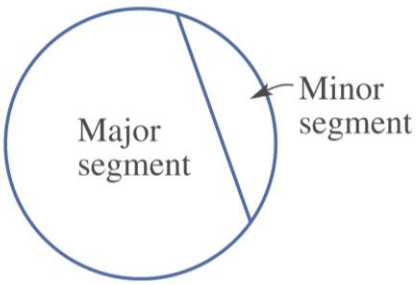
Feature	Definition	Example(s)
Circumference	The circumference of a circle is the perimeter, or distance around the outside of a circle.	
Arc	An arc is a part of the circumference.	
Centre	All points on the circumference is the same distance from the centre. The real definition of a circle is: a shape made by connecting all points an equal distance from the centre.	
Tangent	A tangent is a straight line that touches the circumference at a single point without entering the circle.	
Chord	A chord is a straight line that goes from one point on the circumference to another point	

Diameter	The diameter is a special chord that goes through the centre of the circle.	
Radius	The radius is a line connecting the centre of a circle to the circumference. It is half the diameter.	
Segment	A segment is the area between a chord and the circumference.	
Sector	A sector is the area between two radii.	Quadrant: semicircle

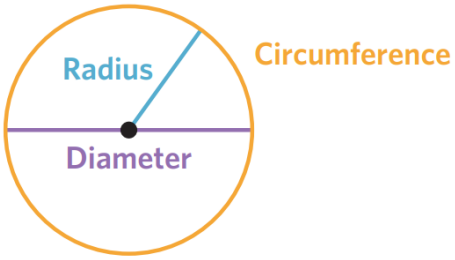
ⓘ Summary of Circle Features







ⓘ Circumference Radius Diameter

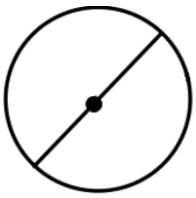


- **Circumference** is a circle's perimeter.
- A **radius** is a straight line from the centre to the circumference.
- A **diameter** is a straight line through the centre, connecting two points on the circumference.
- The radius is half the diameter; the diameter is double the radius.

🔍 **Distinguish** between radius and diameter

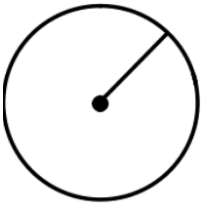
For each circle, identify whether the straight lines show the radius, diameter, or neither.

a



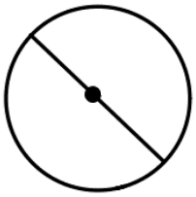
Radius
Diameter
Neither

b



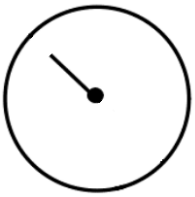
Radius
Diameter
Neither

c



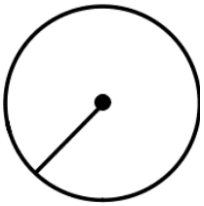
Radius
Diameter
Neither

d



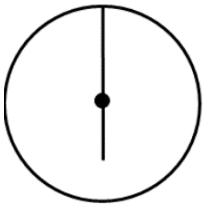
Radius
Diameter
Neither

e



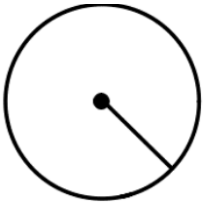
Radius
Diameter
Neither

f



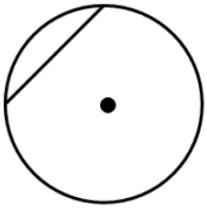
Radius
Diameter
Neither

g



Radius
Diameter
Neither

h



Radius
Diameter
Neither

1 Complete the sentences.

a The circumference is the …… of a circle.

perimeter

b The radius is a straight line connecting the …… to the ……

Centre, circumference

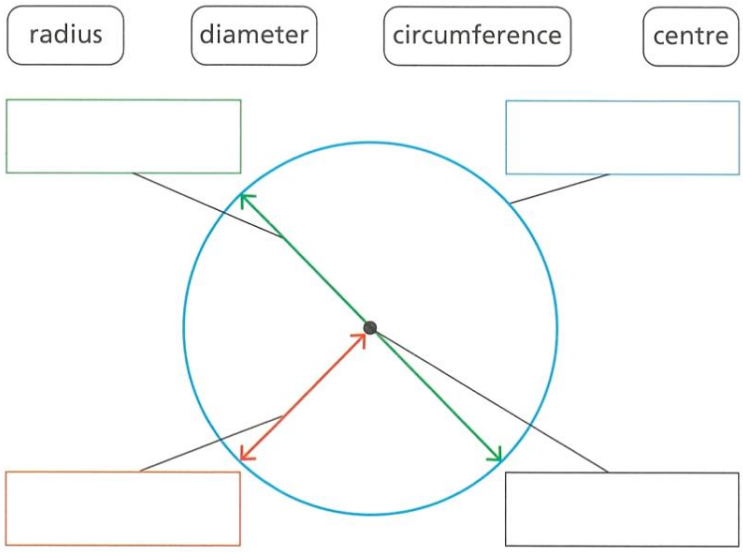
c The diameter is a straight line connecting two points on the ……, passing through the ……

circumference, centre

d The diameter is …… the length of the radius.

double

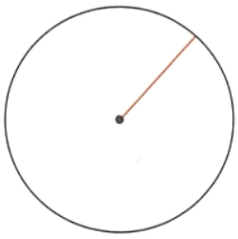
2 Label these parts of the circle.



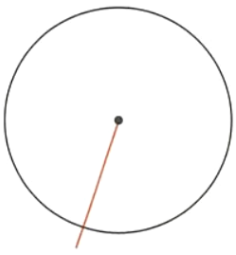
From top left: diameter, circumference, centre, radius

3 Circle which of these (a, b, or c) show a radius.

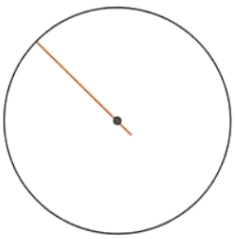
a



b



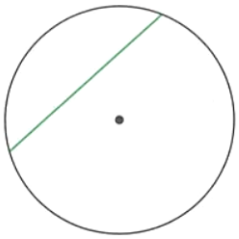
c



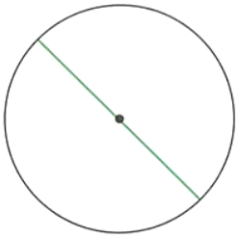
A

4 Circle which of these (a, b, or c) show a diameter.

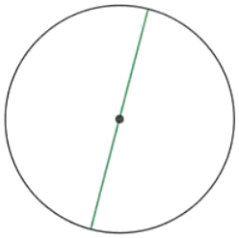
a



b



c



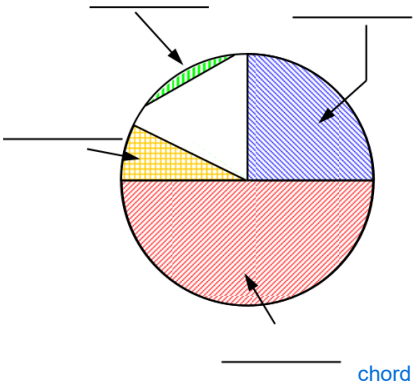
B

5 Complete the table.

Radius	Diameter
4 cm	8 cm
50 mm	100 mm
12.5 cm	25 cm
3.4 cm	6.8 cm
$\frac{3}{4}$ cm	$\frac{6}{4} = \frac{3}{2}$ cm

6 Name the parts of the circle, then complete the sentences.

Answer to diagram, from top left: segment, quadrant, semicircle, sector



chord

segment

Semicircle

a Pizzas are usually sliced into …… sectors

b A …… has a 90-degree angle. quadrant

c Two …… make a semicircle. quadrants

d An arc and two radii make a …… sector

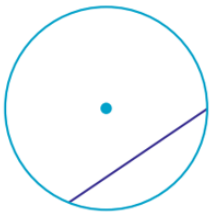
e The …… is a line through the circle that may not touch the centre.

f The area between a chord and the circumference is called the ……

g A …… has the circle's diameter as one of the sides.

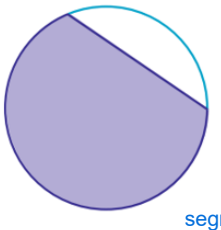
7 Name the highlighted parts of the circle.

a



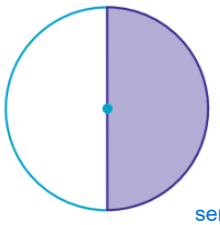
chord

b



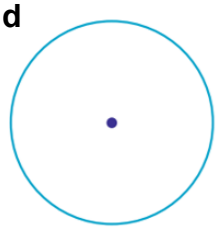
segment

c



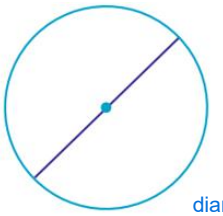
semicircle

d



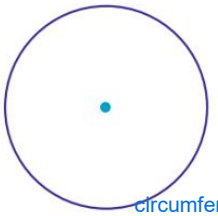
centre

e



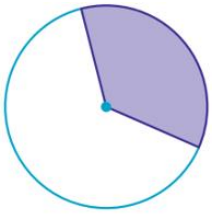
diameter

f



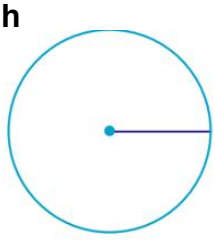
circumference

g



sector

h



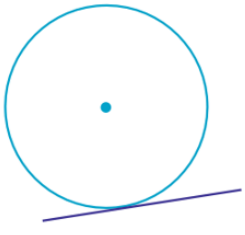
radius

i



segment

j



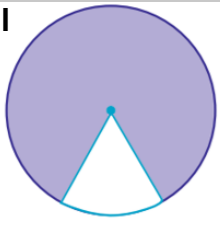
tangent

k



arc

l

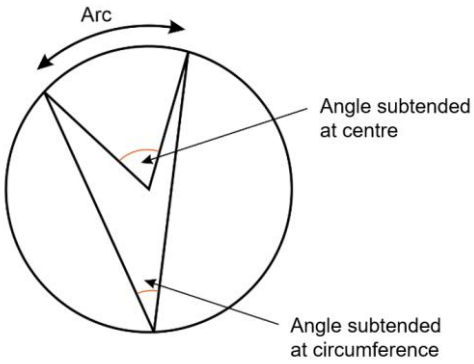


sector

Arcs and Angle at the Centre

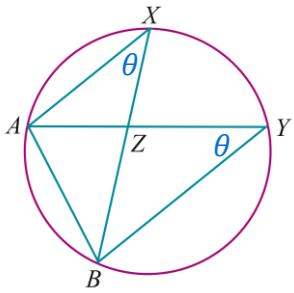
ⓘ Arcs and Angle at the Centre

- An arc/chord **subtends** (or forms) an angle at the centre and the circumference of a circle.
- When an angle is **subtended** by an arc, we say it's **standing** on that arc.



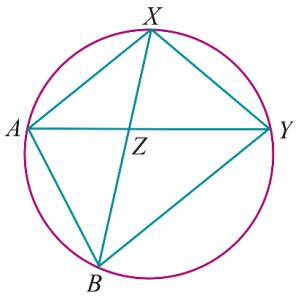
🔍 **Identify** angle subtended by a chord

Refer to the diagram. Label two angles subtended by chord AB at the circumference θ.



$\angle AXB$ and $\angle AYB$

Refer to the diagram.



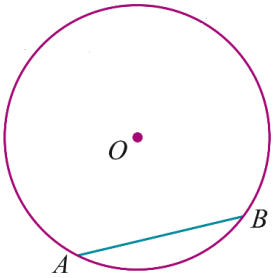
Label two angles θ subtended by chord XY at the circumference.

$\angle XAY$ and $\angle BBY$

Label two angles α subtended by chord AY at the circumference.

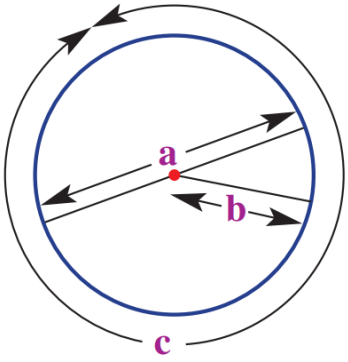
$\angle AXY$ and $\angle ABY$

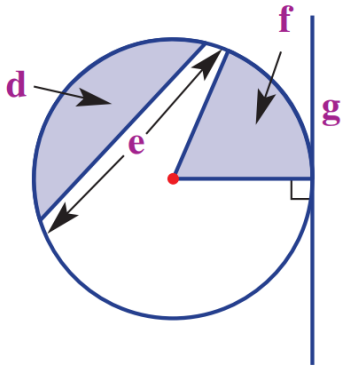
Refer to the diagram. O is the centre of the circle. AB is a chord.



Label the angle θ subtended at the centre by AB.
Draw an angle α subtended at the circumference by AB.

1 Name the features of circles shown.





a diameter

b radius

c circumference

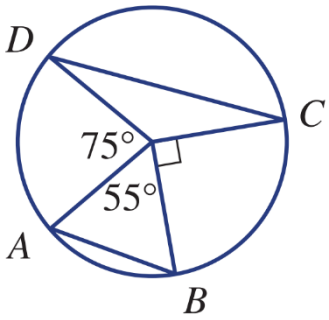
d segment

e chord

f sector

g tangent

2 Give the size of the angle in this circle subtended by the following:



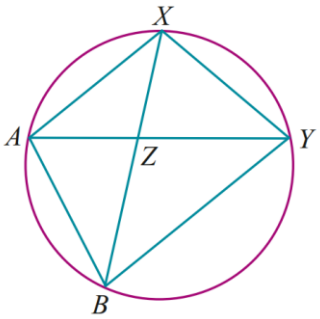
a Chord AB — 55°

b Minor arc BC — 90°

c Minor arc AD — 75°

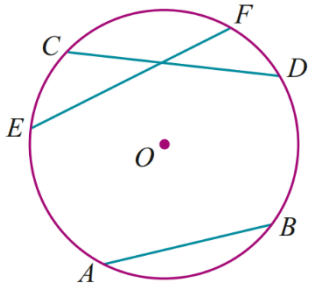
d Chord DC — 140°

3 Name two angles subtended by chord *BX* at the circumference.



$\angle XAB$ and $\angle XYB$

4 O is the centre of the circle.
Draw and name the angle subtended at the centre by each of these chords.



a CD — $\angle COD$

b EF — $\angle EOF$

5 Use each of the following words to fill in the sentences below. Some of the words may be used more than once.

diameter radius circumference chord tangent arc
sector major segment minor segment radii

a A line that goes from the centre of the circle to the edge is called ……

radius

b The distance around the edge of a circle is called the ……

circumference

c Any line that touches the circumference of a circle on the outside once is called a ……

tangent

d A …… is a line that cuts the circle into 2 segments: the …… …… and the …… ……

chord, major segment, minor segment

e An …… is the distance along the circumference between two points.

arc

f 2 radii cut out a …… of the circle

sector

g The …… is twice the ……

Diameter, radius

h A …… is a …… if it passes through the centre of the circle.

Chord, diameter

i The area of the …… plus the area of the …… is equal to the area of the whole circle.

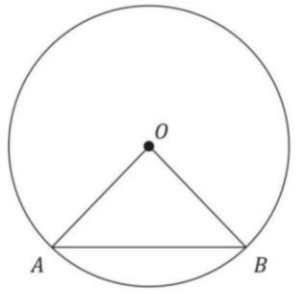
Minor segment, major segment

j A line that connects one point on the circumference to another is called a ……

Chord

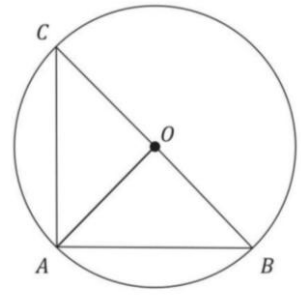
6 Identify the isosceles triangles in the diagrams below.

a



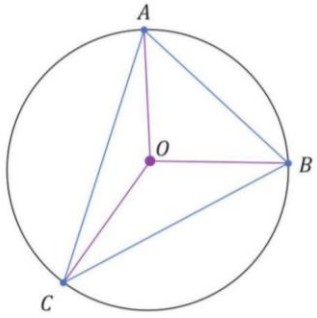
ΔAOB

b



ΔAOB and ΔCOA

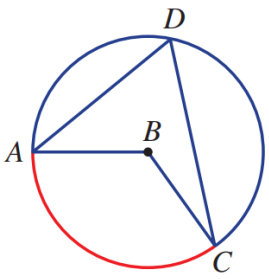
c



$\Delta AOB, \Delta BOC, \Delta AOC$

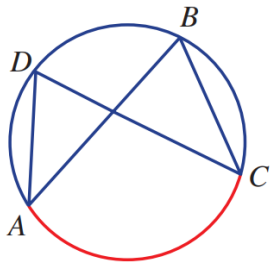
7 Name two angles subtended by the same arc AC in these circles.

a



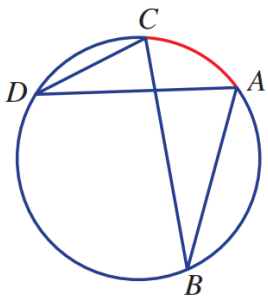
$\angle ABC, \angle ADC$

b



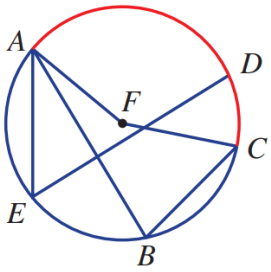
$\angle ADC, \angle ABC$

c



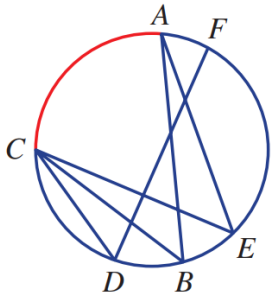
$\angle ABC, \angle ADC$

d



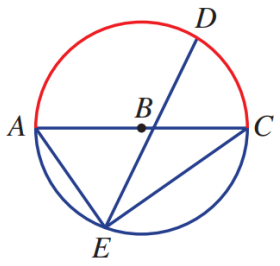
$\angle AFC, \angle ABC$

e



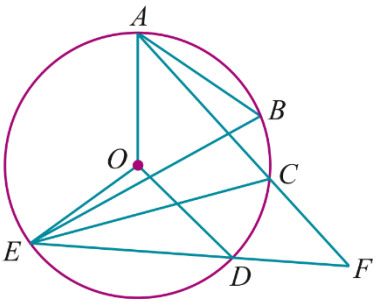
$\angle AEC, \angle ABC$

f



$\angle ABC, \angle AEC$

8 Name the chord that subtends each of these angles.



a $\angle EOD$ — ED

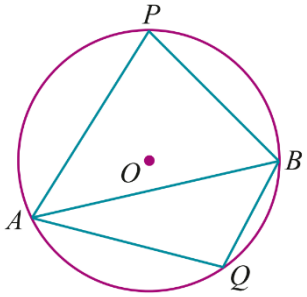
b $\angle CAB$ — CB

c $\angle ABC$ — AC

d $\angle EFA$ — AE

e $\angle EDC$ — EC

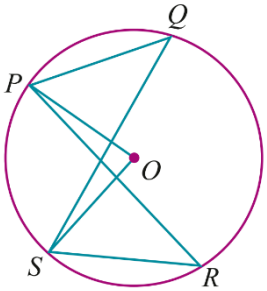
9 Name the angle subtended by chord AB at the circumference in:



a The major segment — $\angle APB$

b The minor segment — $\angle AQB$

10



a Name the angle at the centre subtended by the arc PS.

$\angle POS$

b Name an angle at the circumference on the arc PS.

$\angle PQS$ or $\angle PRS$

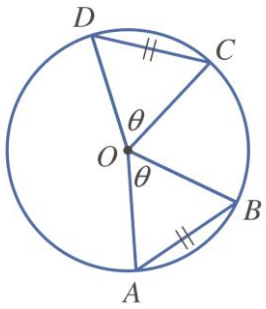
c Name two angles standing on the arc QR.

$\angle QPR$ and $\angle QSR$

Chord Properties

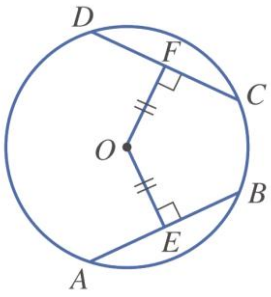
ⓘ Chord Properties

Equal chords subtend equal angles at the centre.



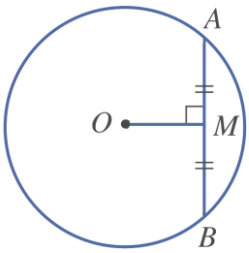
If AB = CD,
then ∠AOB = ∠COD

Equal chords are equidistant from the centre.



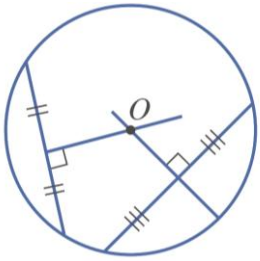
If AB = CD,
then OE = OF

The perpendicular from the centre of a circle bisects the chord.

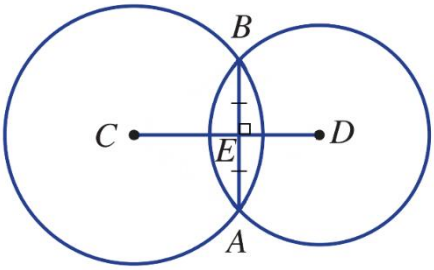


If OM ⊥ AB,
then AM = BM

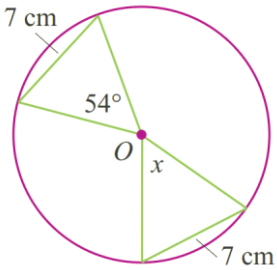
The perpendicular bisector of a chord passes through the centre.



When 2 circles intersect, the line joining their centres bisects their common chord at right angles.

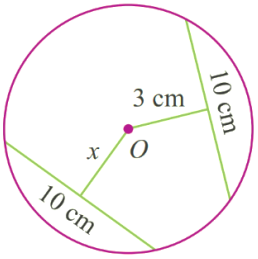


Example Apply chord properties to find unknown values



$x = 54°$

(Equal chords subtend equal angles at centre)

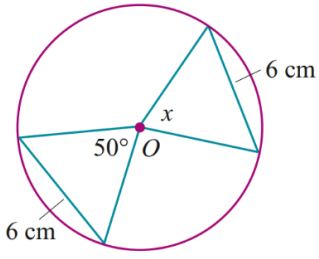


$x = 3$ cm

(Equal chords equidistant from centre)

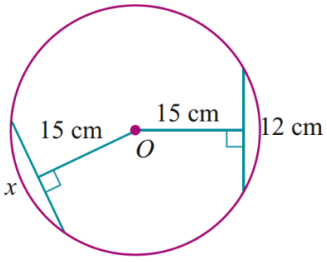
Guided Practice

a



50°

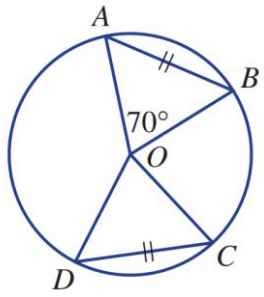
b



12 cm

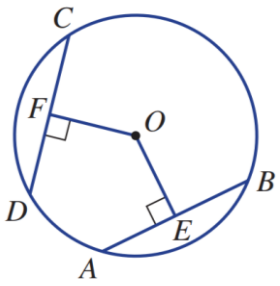
1 For each part, find the required information. You do not need to give reasoning.

a Given AB = CD and ∠AOB = 70°, find ∠DOC.



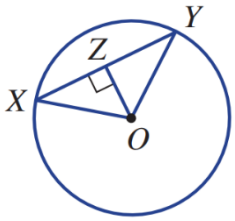
70°

b Given AB = CD and OF = 7.2 cm, find OE.



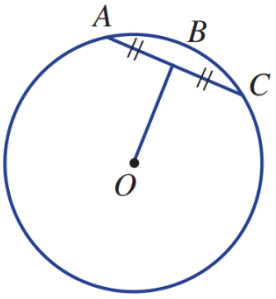
7.2 cm

c Given OZ ⊥ XY, XY = 8 cm and ∠XOY = 102°, find XZ and ∠XOZ.



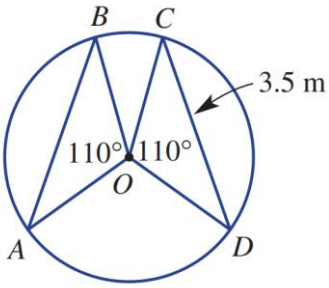
$XZ = 4$ cm
$\angle XOZ = 51°$

d Given AB = BC, find ∠OBC.



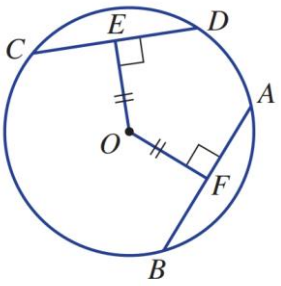
90°

e Given ∠AOB = ∠COD and CD = 3.5 m, find AB.



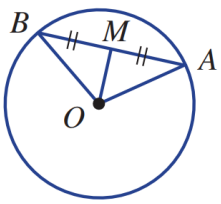
3.5 m

f Given OE = OF and AB = 9 m , find CD.



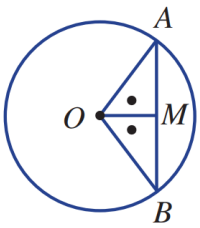
9 m

g Given M is the midpoint of AB, find ∠OMB.



90°

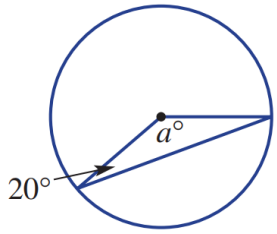
h Given ∠AOM = ∠BOM , find ∠OMB.



90°

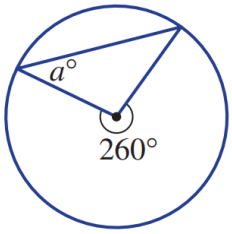
2 Find the size of each unknown value.

a



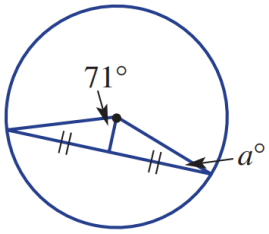
140°

b



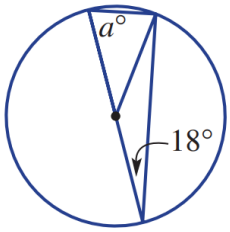
40°

c



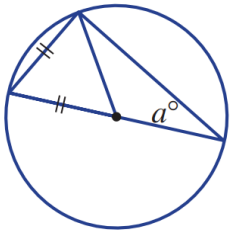
19°

d



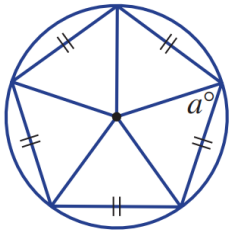
72°

e



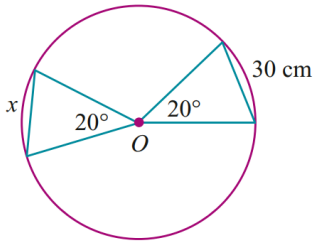
30°

f



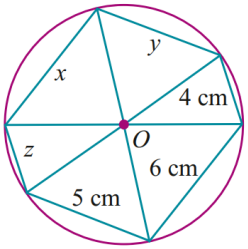
54°

g



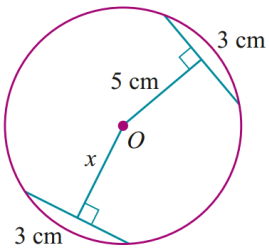
30 cm

h



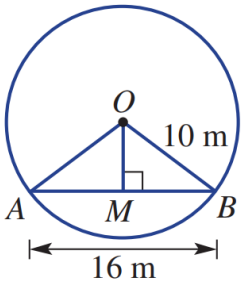
$x = 6, y = 5, z = 4$ cm

i



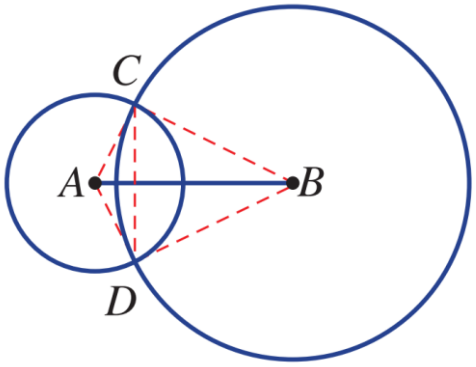
$x = 5$ cm

3 Find the length OM. Use Pythagoras' Theorem.



6 m

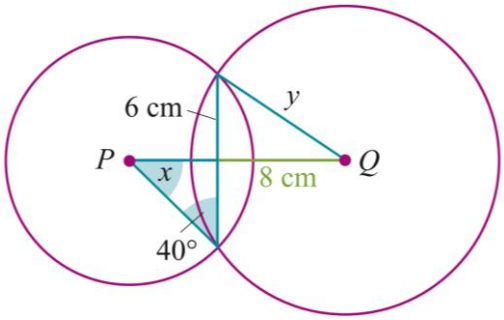
4 In this diagram, radius AD = 5 mm, radius BD = 12 mm and chord CD = 8 mm.
Find the exact length of AB, in surd form.



$3 + 8\sqrt{2}$ mm

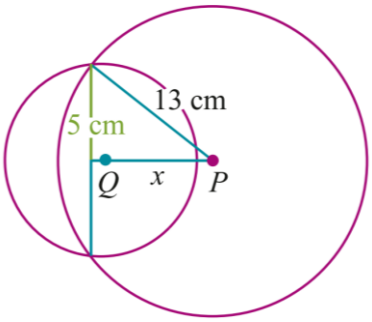
5 In the diagrams below, P and Q are the centres of the circles. Find the values of the variables.

a



$x = 50°, y = 10$ cm

b



$x = 12$ cm

6 Complete the following proofs.

a *Proof:* In $\triangle AOB$ and $\triangle COD$:

$AB = AC$	(given)
$AO = CO$	(radii)
$BO = DO$	(radii)
$\therefore \triangle AOB \equiv \triangle COD$	(SSS)
Hence $\angle AOB = \angle COD$	(Matching angles in congruent $\triangle$s are equal.)
Also $\angle ABO = \angle CDO$	(Matching angles in congruent $\triangle$s are equal.)

b *Proof:* In $\triangle OPB$ and $\triangle OQD$:

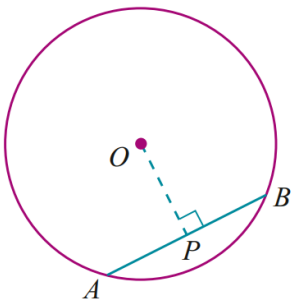
$\angle OPB = \angle OQD = 90°$	(given)
$\angle OBP = \angle ODQ$	(from part **a**)
$BO = DO$	(radii)
$\therefore \triangle OPB \equiv \triangle OQD$	(AAS)
$\therefore OP = OQ$	(Matching sides in congruent $\triangle$s are equal.)

From parts **a** and **b** we can conclude that:

Chords of equal length in a circle subtend equal angles at the centre and are equidistant from the centre.

c Use the diagram to prove △OAP ≡ △OBP, hence prove:

The perpendicular drawn from the centre of a circle to a chord bisects the chord.



$\angle OPA = \angle OPB = 90°$

$OA = OB$ (radii of a circle)

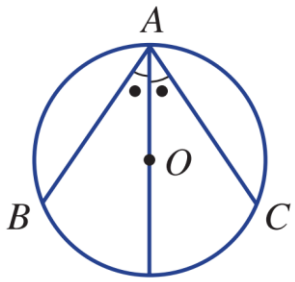
OP is common

△OAP ≡ △OBP (RHS)

$\therefore AP = PB$ (corresponding sides of congruent triangles equal)

$\therefore$ The perpendicular drawn from the centre of a circle to a chord bisects the chord.

5 In this circle ∠BAO = ∠CAO. Prove AB = AC. Suggestion: Construct BC.

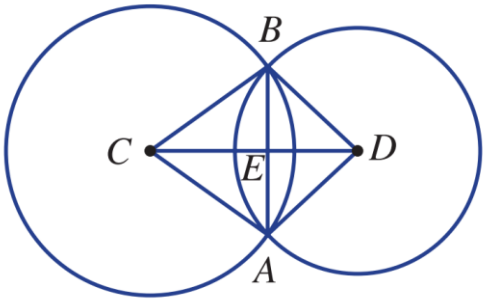


First, prove △OAB ≡ △OAC (AAS), which are isosceles.

So, AB = AC, corresponding sides in congruent triangles.

6 In this question we will prove the following theorem. When two circles intersect, the line joining their centres bisects that common chord at right angles.

a Prove △ACD ≡ △BCD.



AD = BD (equal radii),

AC = BC (equal radii),

CD is common.

Therefore, △ACD ≡ △BCD (SSS).

b Hence, prove △ACE ≡ △BCE.

AC = BC,

∠ACE = ∠BCE (corresponding angles),

CE is common.

Therefore, △ACE ≡ △BCE (SAS).

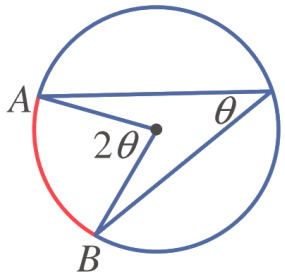
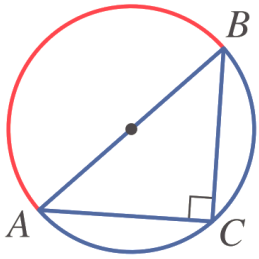
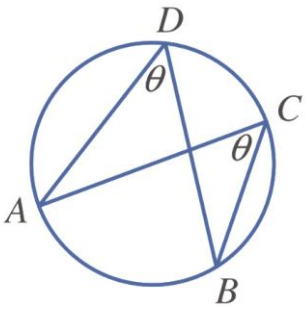
c Hence, prove CD ⊥ AB and BE = EA.

∠BEC = ∠CEA = 90° (equal angles on a straight line).

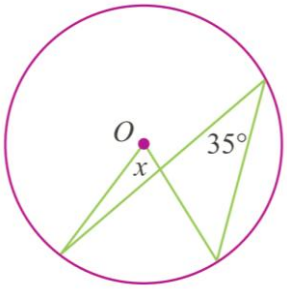
Therefore, CD ⊥ AB and BE = EA (matching sides in congruent triangles).

Angle Properties

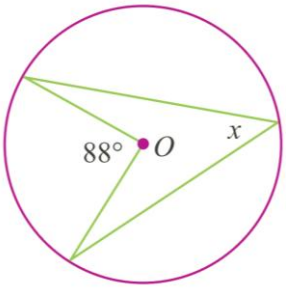
ⓘ Angle Properties

The angle at the centre is twice the angle at the circumference when standing on the same arc.	The angle in a semicircle is 90°	Angles at the circumference standing on the same arc are equal.
		

Example Apply angle properties to find unknown values



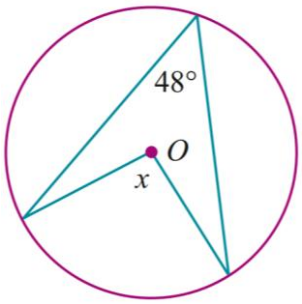
$x = 70°$ (Angle at centre twice angle at circumference when on the same arc.)



$x = 44°$ (angle at centre is twice angle at circumference when on the same arc.)

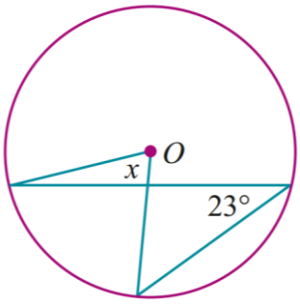
Guided Practice

a



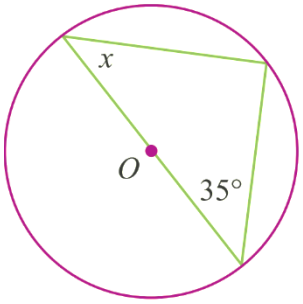
96

b



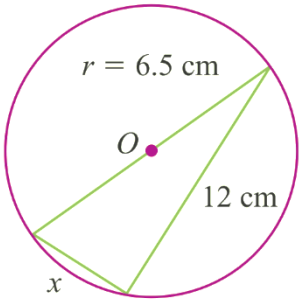
46

c



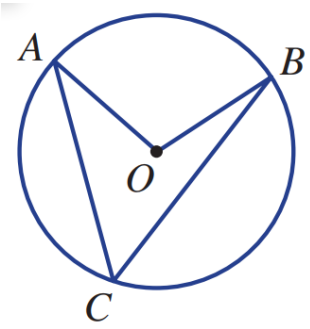
55

d



5 cm

1 For this circle, O is the centre.



a Name the angle at the centre of the circle subtended by AB.

∠AOB

b Name the angle at the circumference subtended by AB.

∠ACB

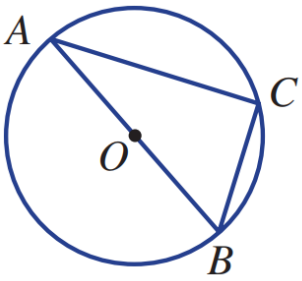
c If ∠ACB = 40°, find ∠AOB.

80°

d If ∠AOB = 122°, find ∠ACB.

61°

2 For this circle, AB is a diameter.



a What is the size of ∠AOB?

180°

b What is the size of ∠ACB?

90°

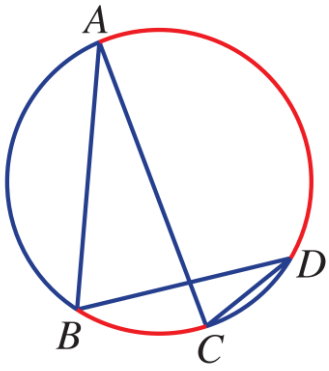
c If ∠CAB = 30°, find ∠ABC.

60°

d If ∠ABC = 83°, find ∠CAB.

7°

3 For this circle, answer the following.



a Name two angles subtended by arc AD.

∠ABD and ∠ACD

b State the size of ∠ACD if ∠ABD = 85°.

85°

c Name two angles subtended by arc BC.

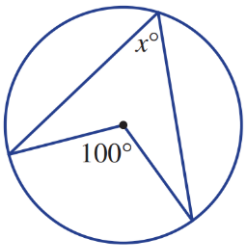
∠BAC and ∠BDC

d State the size of ∠BAC if ∠BDC = 17°.

17°

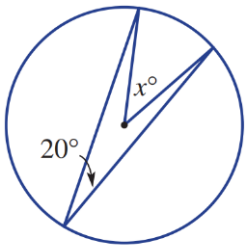
4 Find the value of x in these circles.

a



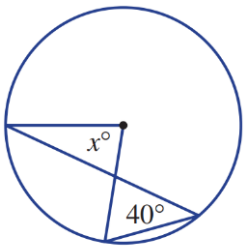
50°

b



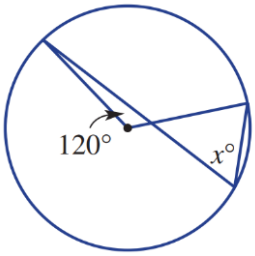
40°

c



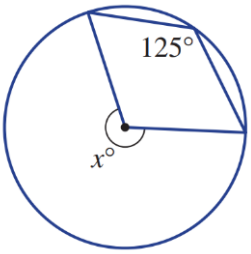
80°

d



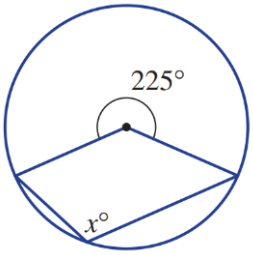
60°

e



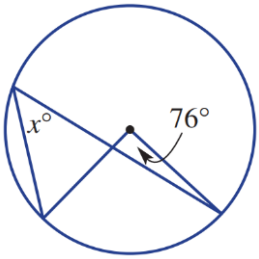
250°

f



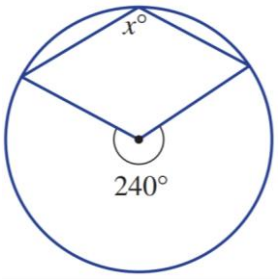
112.5°

g



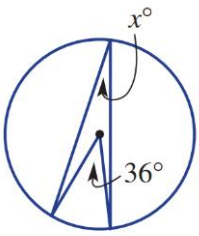
38°

h



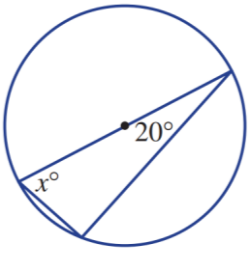
120°

i



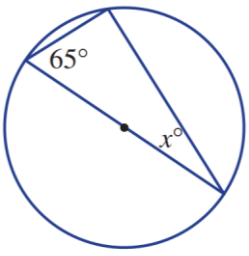
18°

j



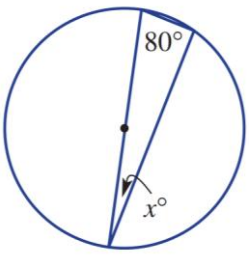
70°

k



25°

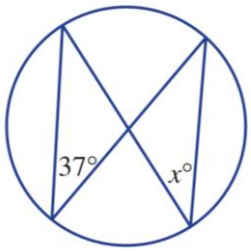
l



10°

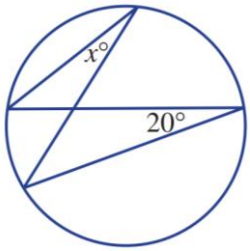
5 Find the value of x.

a



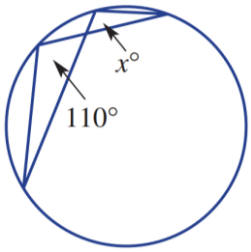
37°

b



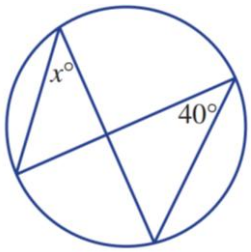
20°

c



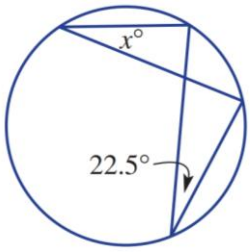
110°

d



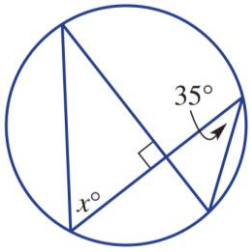
40°

e



22.5°

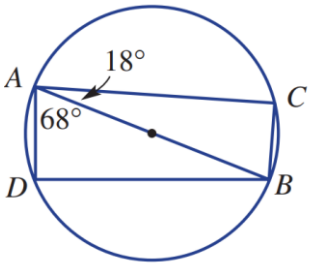
f



55°

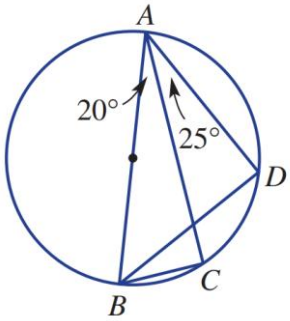
6 Find the size of both ∠ABC and ∠ABD.

a



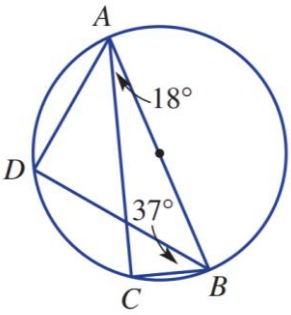
∠ABC = 72°, ∠ABD = 22°

b



∠ABC = 70°, ∠ABD = 45°

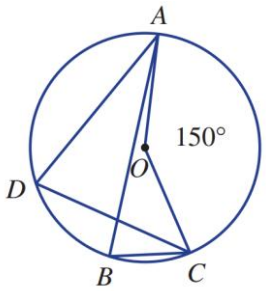
c



∠ABC = 72°, ∠ABD = 35°

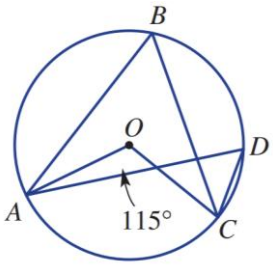
7

a Find ∠ADC and ∠ABC



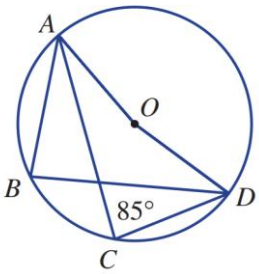
∠ADC = 75°, ∠ABC = 75°

b Find ∠ABC and ∠ADC



∠ABC = 57.5°, ∠ADC = 57.5°

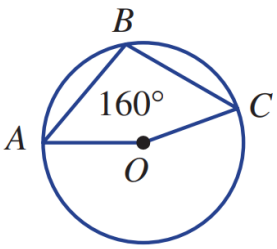
c Find ∠AOD and ∠ABD



∠AOD = 170°, ∠ABD = 85°

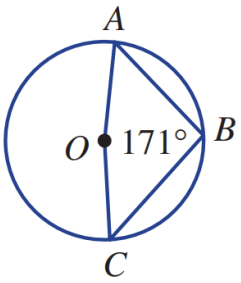
8 Find ∠ABC.

a



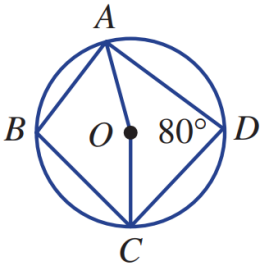
100°

b



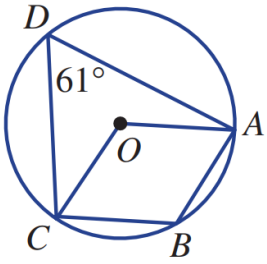
94.5°

c



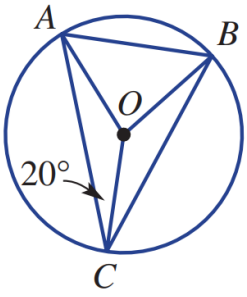
100°

d



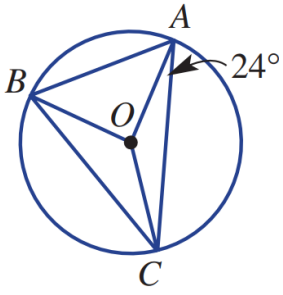
119°

e



70°

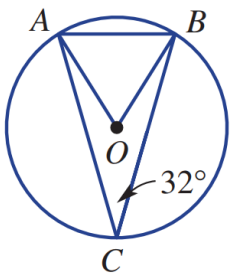
f



66°

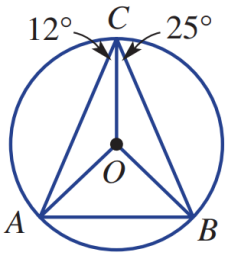
9 Find ∠OAB.

a



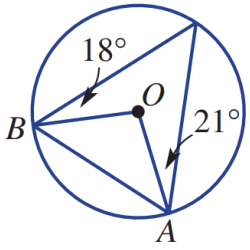
58°

b



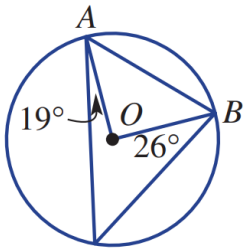
53°

c



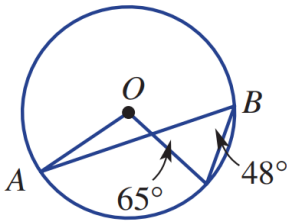
51°

d



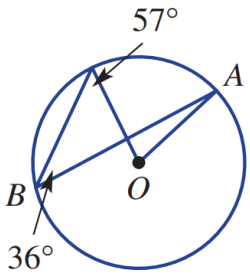
45°

e



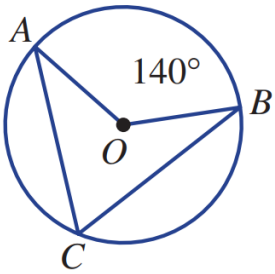
19°

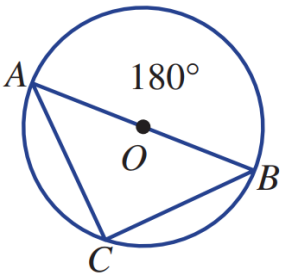
f



21°

10





a For the first circle shown, find ∠ACB

70°

b For the second circle shown, find ∠ACB

90°

c Explain why “The angle in a semicircle is 90°” is a special case of “The angle at the centre of a circle is twice the angle at the circumference when standing on the same arc.”

The second circle is the specific case of the first circle when the angle at the centre is 180°.

11 Complete the following proofs



a *Proof:* Let $\angle ACO = x$ and $\angle OCB = y$.
In $\triangle AOC$:
$\angle OAC = \angle OCA$ (Base angles of an isosceles $\triangle$ are equal; equal radii.)
$= x$
Now $\angle POA = \angle OAC + \angle OCA$
(The exterior angle of a $\triangle$ equals the sum of the interior opposite angles.)
$= 2x$
Similarly $\angle POB = 2y$
$\therefore \angle AOB = 2x + 2y = 2(x + y)$
$= 2 \times \angle ACB$

b
Proof: Let $\angle ACO = x$ and $\angle OCB = y$.
Show that $\angle POA = 2x$ and $\angle POB = 2y$.
Now $\angle AOB = \angle POB - \angle POA$
$= 2y - 2x = 2(y - x)$
But $\angle ACB = y - x$
$\therefore \angle AOB = 2 \times \angle ACB$

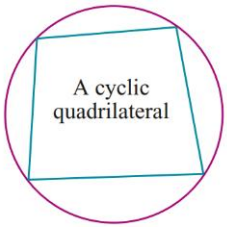
c *Proof:* $\angle APB = \frac{1}{2} \times \angle AOB$ and
(The angle at the centre of a circle is double the angle at the circumference, standing on the same arc.)
$\angle AQB = \frac{1}{2} \times \angle AOB$
(The angle at the centre of a circle is double the angle at the circumference, standing on the same arc.)
$\therefore \angle APB = \angle AQB$

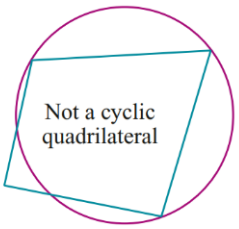
d *Proof:* $\angle AOB = 2 \times \angle ACB$
(The angle at the centre of a circle is double the angle at the circumference, standing on the same arc.)
But $\angle AOB = 180°$ (Angles on a straight line equal 180°.)
$2 \times \angle ACB = 180°$
$\therefore \angle ACB = 90°$

Cyclic Quadrilaterals

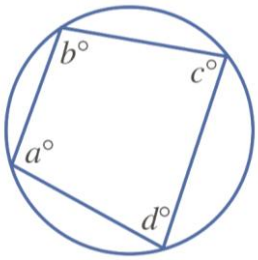
ⓘ Cyclic Quadrilaterals

- A **cyclic quadrilateral** is a four-sided shape whose vertices all lie on a circle.





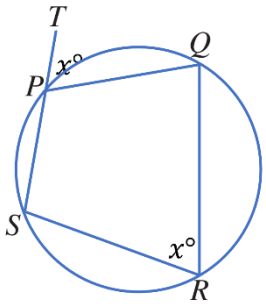
The opposite angles of cyclic quadrilaterals sum to 180° (supplementary).



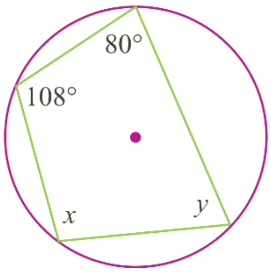
$a + c = 180°$

$b + d = 180°$

Exterior angles in cyclic quadrilaterals equal the opposite interior angle



Example Apply properties of cyclic quadrilaterals to find unknown values

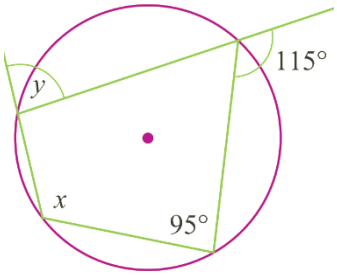


$x + 80° = 180°$

$x = 100°$

$108° + y = 180°$

$y = 72°$

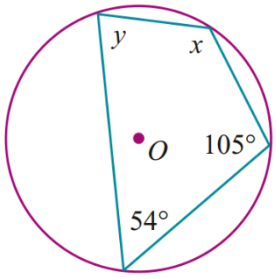


$x = 115°$

$y = 95°$

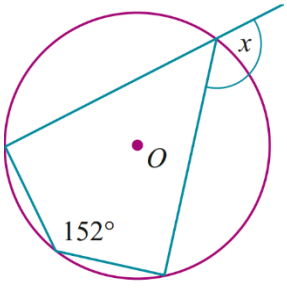
Guided Practice

a



$x = 126, y = 75$

b

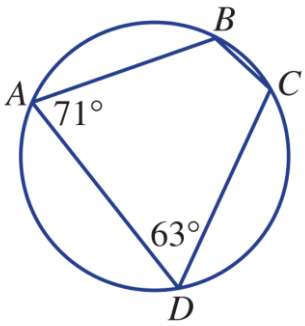


$x = 152$

1 Opposite angles in a cyclic quadrilateral are supplementary.

a What does it mean when we say two angles are supplementary?

Supplementary angles sum to 180°.



b Find ∠ABC.

117°

c Find ∠BCD.

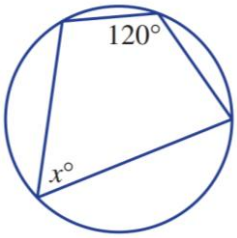
109°

d Check that ∠ABC + ∠BCD + ∠CDA + ∠DAB = 360°.

117° + 109° + 63° + 71° = 360

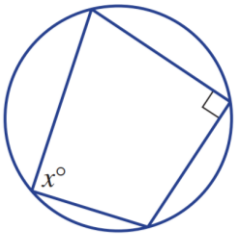
2 Find the value of the pronumerals in these circles.

a



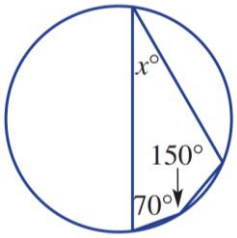
60°

b



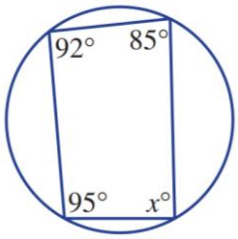
90°

c



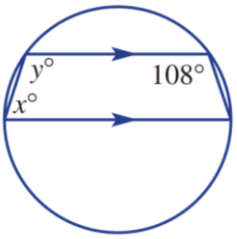
30°

d



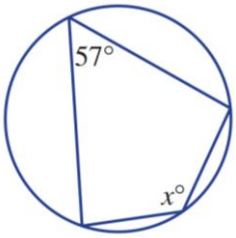
88°

e



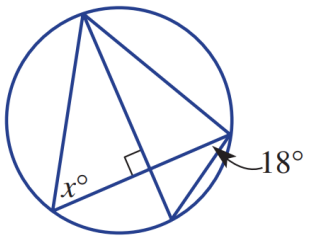
$x = 72°, y = 108°$

f



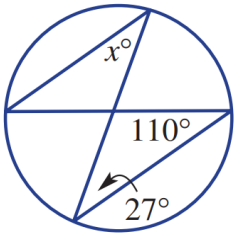
123°

g



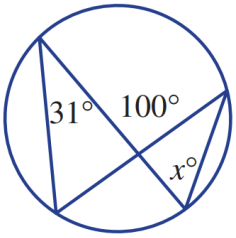
72

h



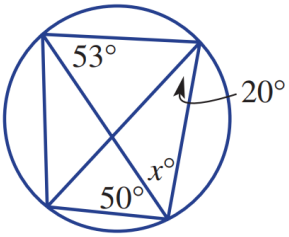
43

i



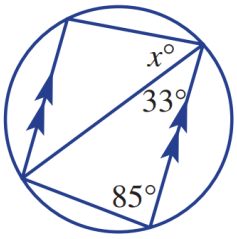
69

j



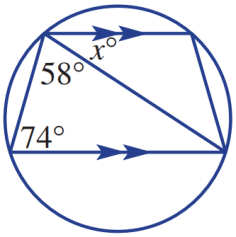
57

k



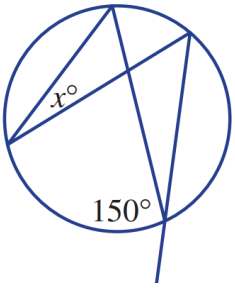
52

l



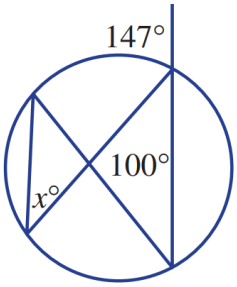
48

m



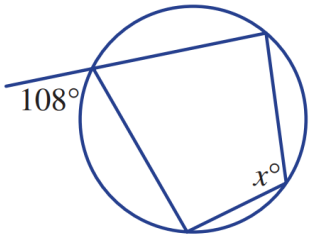
30

n



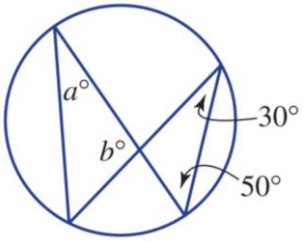
47

o



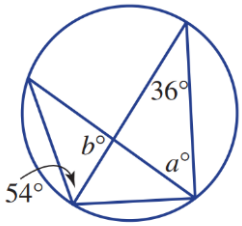
108

p



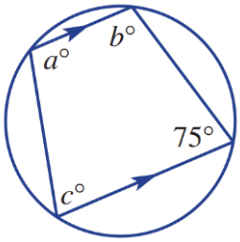
a = 30, b = 100

q



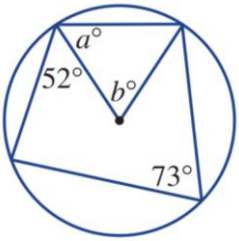
a = 54, b = 90

r



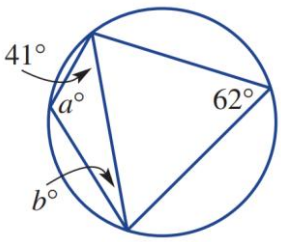
a = 105, b = 105, c = 75

s



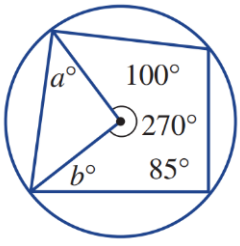
a = 55, b = 70

t



a = 118, b = 21

u



a = 45, b = 35

3 *Proof:* Let $\angle ABP = x$

$\angle ABC = 180° - x$ (CBP is a straight line.)

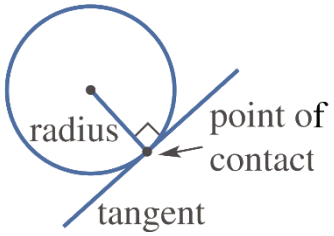
$\angle ADC = x$ (Opposite angles in a cyclic quadrilateral are supplementary.)

$\therefore \angle ABP = \angle ADC.$

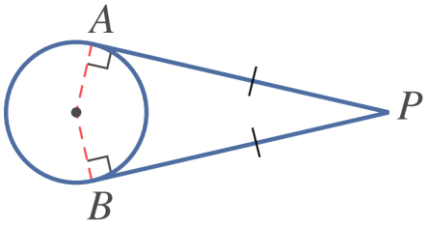
Tangent and Secant Properties

ⓘ Tangent and Secant Properties

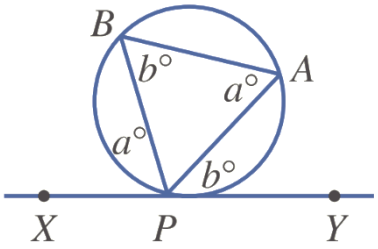
Angle between tangent and radius = 90°



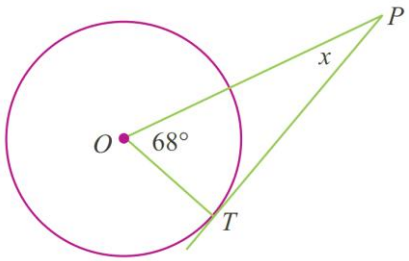
Tangents from an external point are equal in length.



Alternate segment theorem
The angle between a tangent and a chord at the point of contact is equal to the angle in the alternate segment.

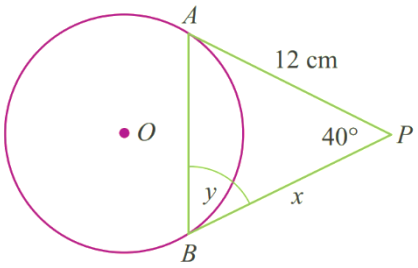


Example Apply tangent and secant properties to find unknown values



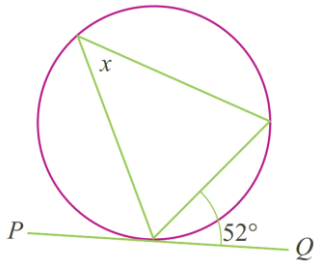
$\angle OTP = 90°$
(A tangent is perpendicular to the radius at the point of contact)

$x + 90° + 68° = 180°$
$x = 22°$



$x = 12$ cm
(Tangents drawn to a circle from an external point are equal.)

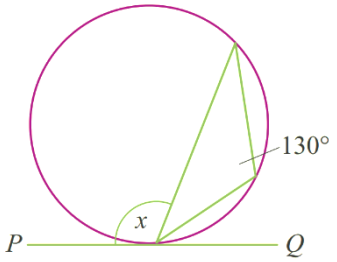
$y = \frac{180 - 40}{2} = 70°$
(Base angles of isosceles triangle equal)



$x = 52°$
(Alternate segment theorem)

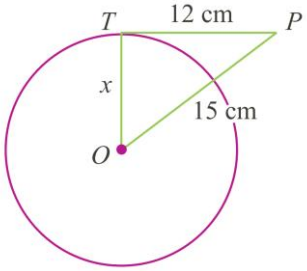
Guided Practice

a



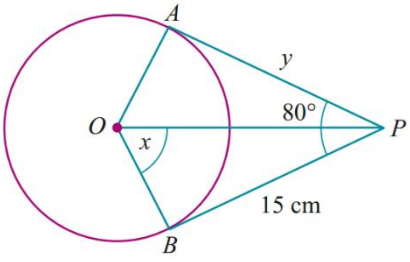
130°

b



9 cm

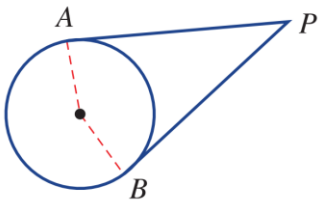
c



$x = 50°, y = 15°$

1

a How many times does a tangent intersect a circle?



once

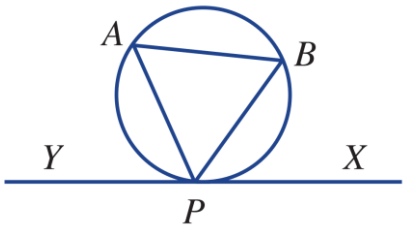
b At the point of contact, what angle does the tangent make with the radius?

90°

c If AP is 5 cm, what is the length of BP in this diagram?

Also 5 cm

2 For this diagram name the angle that is:



a equal to ∠BPX

∠BAP

b equal to ∠BAP

∠BPX

c equal to ∠APY

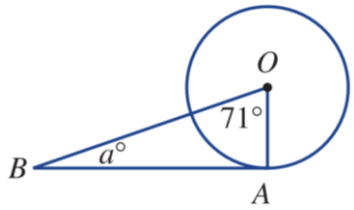
∠ABP

d equal to ∠ABP

∠APY

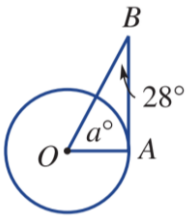
3 Find the value of a in these diagrams that include tangents.

a



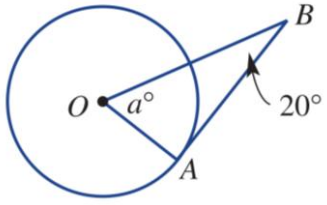
19

b



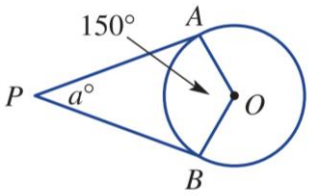
62

c



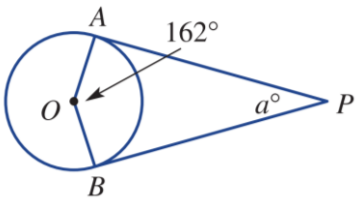
70

d



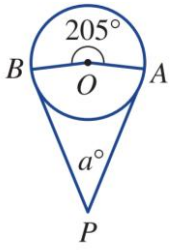
30

e



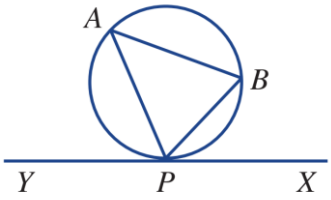
18

f



25

4 In this diagram, XY is a tangent to the circle. Find:



a ∠PAB if ∠BPX = 50

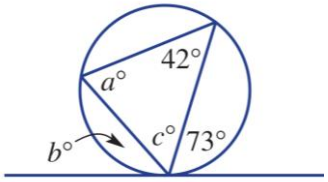
50°

b ∠APY if ∠ABP = 59°

59°

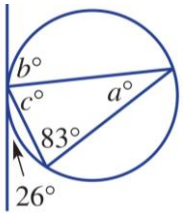
5 Find the value of a, b and c in these diagrams.

a



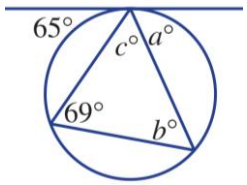
a=73, b=42, c=65

b



a=26, b=83, c=71

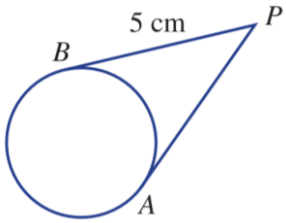
c



a=69, b=65, c=46

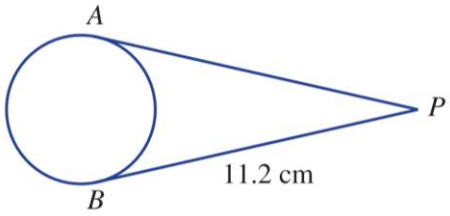
6 Find the length of AP

a



5 cm

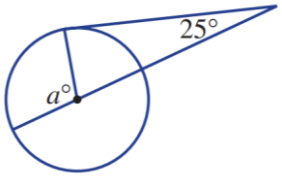
b



11.2 cm

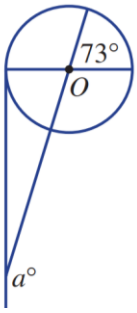
7 Find the value of a

a



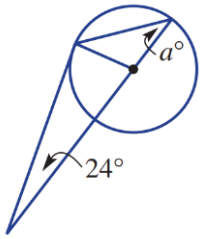
115

b



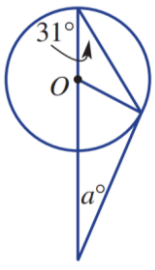
163

c



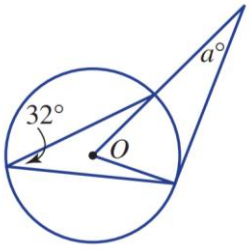
33

d



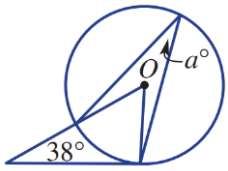
28

e



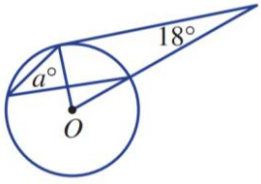
26

f



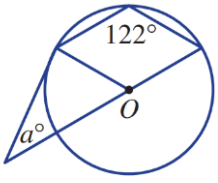
26

g



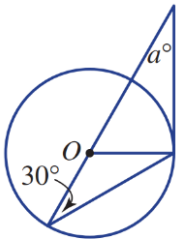
36

h



26

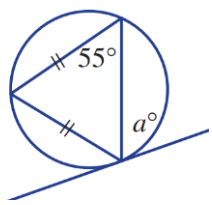
i



30

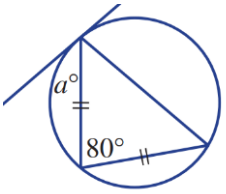
8 Find the value of a.

a



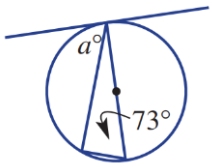
70

b



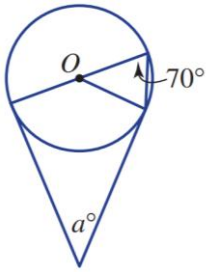
50

c



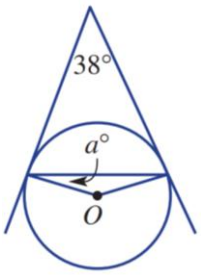
73

d



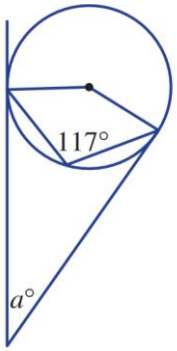
40

e



19

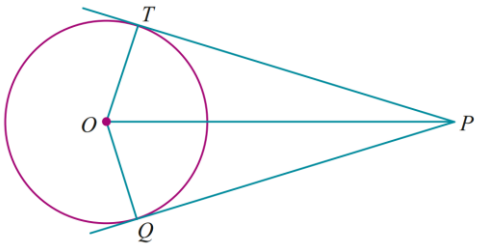
f



54

9 PT and PQ are the tangents from P to a circle with centre O.

a Prove △PTO ≡ △PQO.



$\angle OTP = \angle OQT = 90°$ (tangents meet radii at 90°)

$OT = OQ$ (radii)

OP is common

△ PTO ≡ △ PQO (RHS)

b Hence prove PT = PQ.

$PT = PQ$ (corresponding sides of congruent triangles equal)

Two tangents drawn to a circle from an external point are equal in length.

10

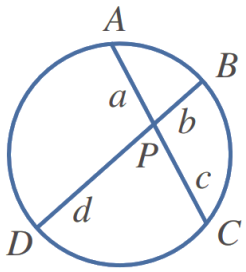
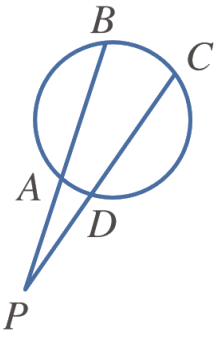
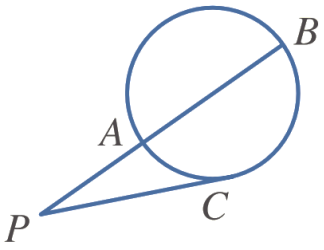
Proof: Join CO and BO. Let $\angle BCQ = x$.

$\angle OCB = 90° - x$	(A tangent to a circle is perpendicular to the radius at the point of contact.)
$\angle OBC = 90° - x$	(Base angles of an isosceles △ are equal)
$\angle COB = 180° - 2x$	(Angle sum of a △ is 180°.)
$\angle BAC = 90° - x$	(The angle at the centre of a circle is double the angle at the circumference, standing on the same arc.)

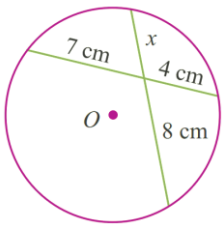
$\therefore \angle BCQ = \angle CAB$

Further Tangent and Secant Properties

ⓘ Further Tangent and Secant Properties

The products of the intercepts of two intersecting chords are equal.	The products of the intercepts of 2 intersecting secants to a circle from an external point are equal.	The square of a tangent to a circle from an external point equals the product of the intercepts of any secants from the point.
		
$ac = bd$	$AP \times BP = DP \times CP$	$CP^2 = AP \times BP$

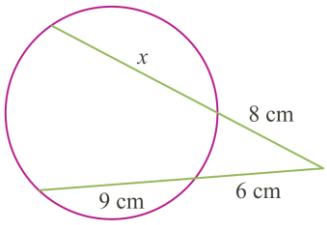
Example Apply tangent and secant properties to find unknown values



$7 \times 4 = 8x$

$28 = 8x$

$\frac{7}{2} = x$

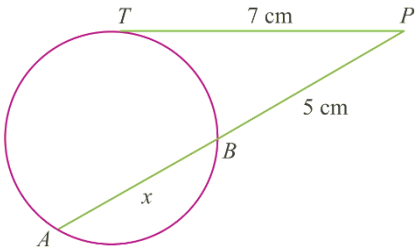


$8(x + 8) = 6(6 + 9)$

$8x + 64 = 90$

$8x = 26$

$x = 3.25$



$7^2 = 5(5 + x)$

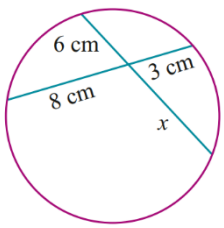
$49 = 25 + 5x$

$24 = 5x$

$4.8 = x$

Guided Practice

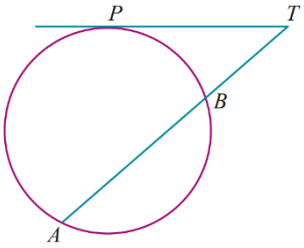
a



4 cm

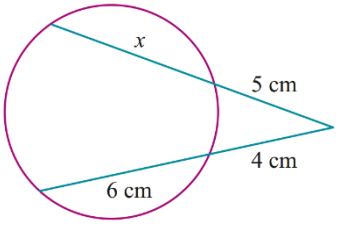
b $TA = 8$ cm, $TB = 2$ cm

Find TP



4 cm

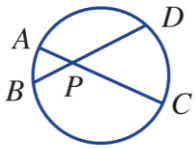
c



3 cm

1 Complete the rules for each diagram.

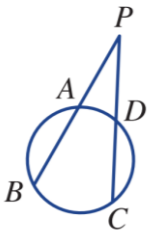
a



$AP \times CP = __ \times __$

AP×CP=BP×DP

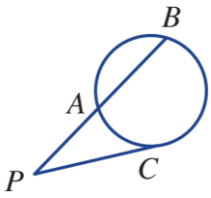
b



$AP \times __ = __ \times __$

AP×BP=DP×CP

c

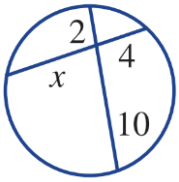


$AP \times __ = __^2$

AP×BP=CP²

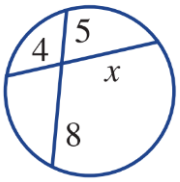
2 Find the length of x in each figure.

a



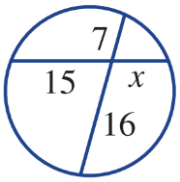
5

b



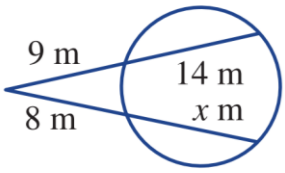
10

c



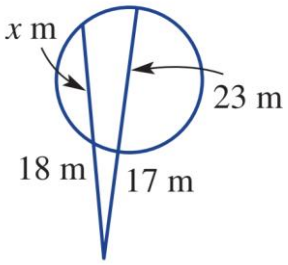
$\frac{112}{15}$

d



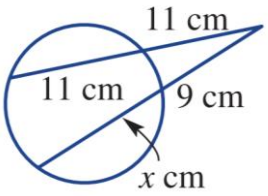
$\frac{143}{8}$

e



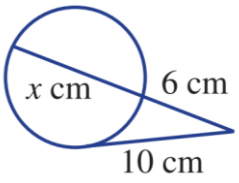
$\frac{178}{9}$

f



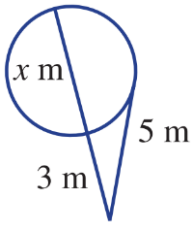
$\frac{161}{9}$

g



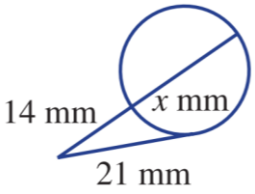
$\frac{32}{3}$

h



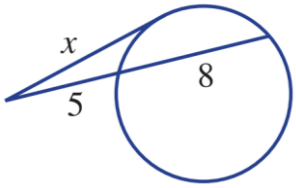
$\frac{16}{3}$

i



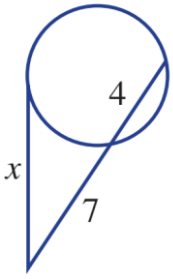
$\frac{35}{2}$

j



$\sqrt{65}$

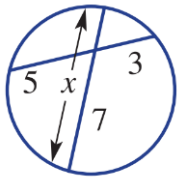
k



$\sqrt{77}$

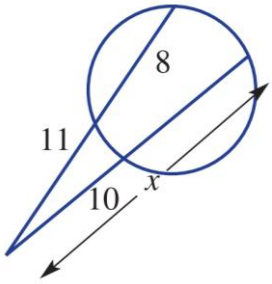
3 Find the value of x.

a



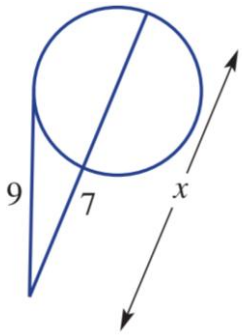
$\frac{64}{7}$

b



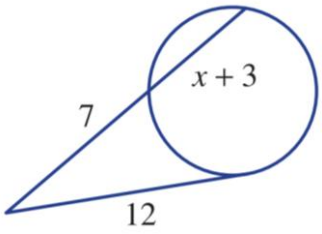
$\frac{209}{10}$

c



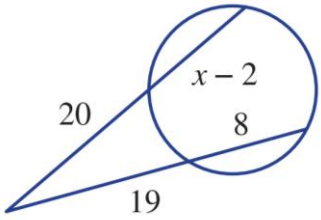
$\frac{81}{7}$

d



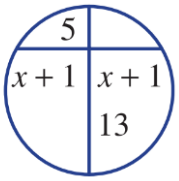
$\frac{74}{7}$

e



$\frac{153}{20}$

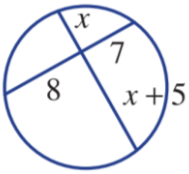
f



$\sqrt{65} - 1$

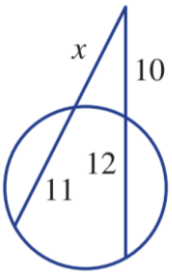
4 For each diagram, derive the equation.

a $x^2 + 5x - 56 = 0$



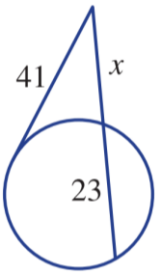
$x(x + 5) = 7 \times 8,$
$x^2 + 5x = 56,$
$x^2 + 5x - 56 = 0$

b $x^2 + 11x - 220 = 0$



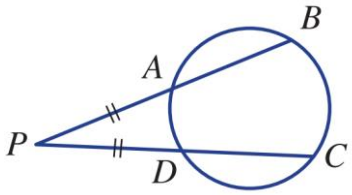
$x(x + 11) = 10 \times 22,$
$x^2 + 11x = 220,$
$x^2 + 11x - 220 = 0$

c $x^2 + 23x - 1681 = 0$



$x(x + 23) = 412,$
$x^2 + 23x = 1681,$
$x^2 + 23x - 1681 = 0$

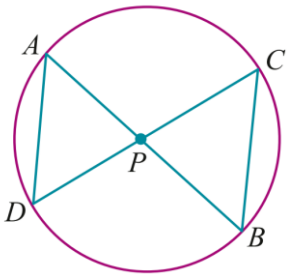
5 In this diagram AP = DP. Explain why AB = DC.



$AP^2 = DP \times CP$ and $BP^2 = DP \times CP$,
so BP=AP.

6 AB and CD are two chords of a circle that intersect at P.

a Prove that △APD is similar to △CPB



$\angle DAB = \angle DCB$ (angles subtended at circumference by chord DB equal)

$\angle ADC = \angle ABC$ (angles subtended at circumference by chord AC equal)

$\angle ADP = \angle BCP$ (vertically opposite)

$\therefore$△ APD ||| △ CPB (equiangular)

b Write down all the equal ratios of sides in these two triangles.

$\frac{AP}{PC} = \frac{DP}{PB} = \frac{AD}{CB}$

c Hence show that AP × PB = DP × PC

Cross multiply first two fractions from above.

7

Proof: In $\triangle PTB$ and $\triangle ATP$:

$\angle TPB = \angle PAB$ (The angle between a tangent and a chord is equal to the angle in the alternate segment.)

$\angle PTB = \angle ATP$ (common)

$\triangle PTB$ is similar to $\triangle ATP$. (Two angles of one △ equal two angles of the other △.)

$\frac{AP}{PB} = \frac{TP}{TB} = \frac{TA}{TP}$

$\therefore\ TP^2 = TA \times TB.$

8 *Proof:* $PT^2 = TB \times TA$ and $PT^2 = TD \times TC$

(The square of the length of a tangent from an external point equals the product of the intercepts of any secant from the point.)

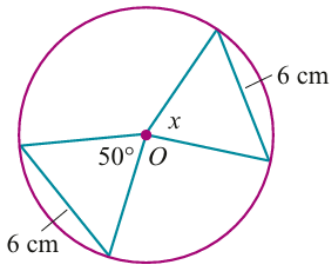
$\therefore\ TB \times TA = TD \times TC$

Mixed Practice

FOUNDATION

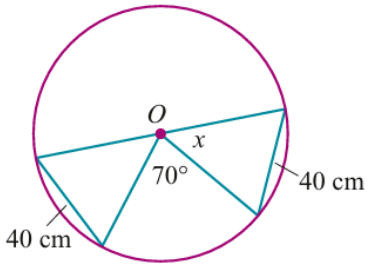
1 Find the value of the pronumeral. You do not need to give reasoning.

a



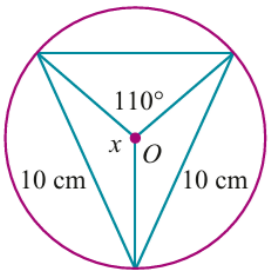
50°

b



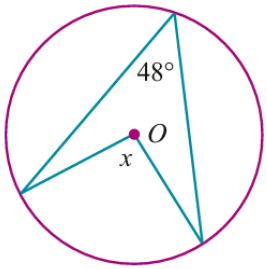
55°

c



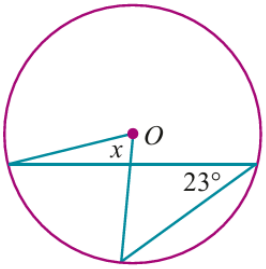
125°

d



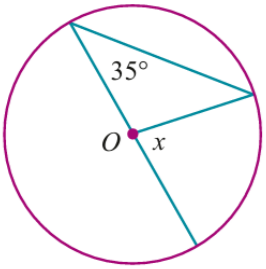
96°

e



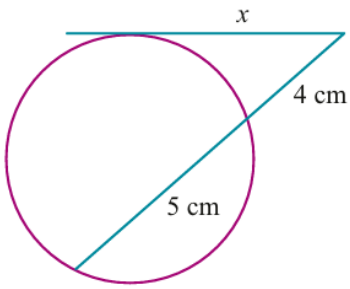
46°

f



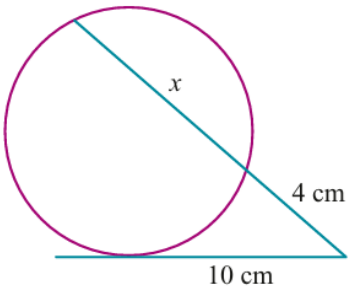
70°

g



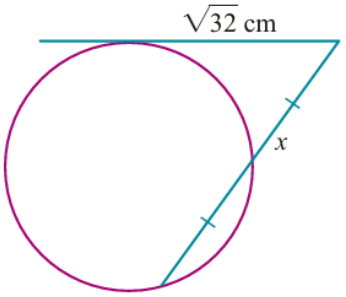
6 cm

h



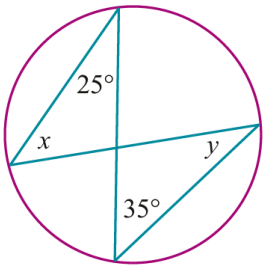
21 cm

i



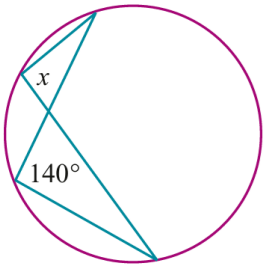
4 cm

j



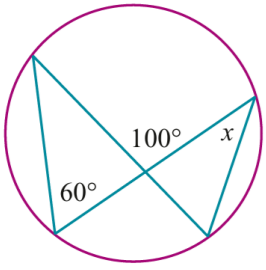
$x = 35°, y = 25°$

k



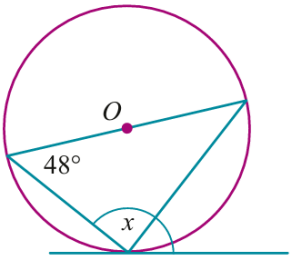
140°

l



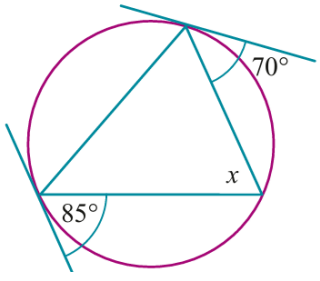
40°

m



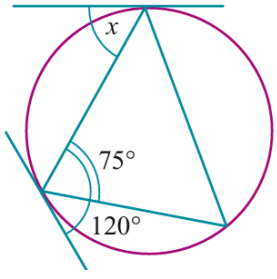
138°

n



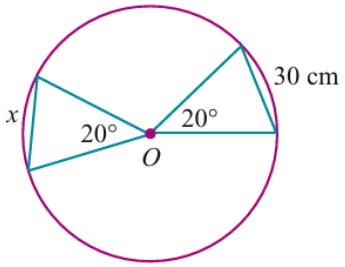
25°

o



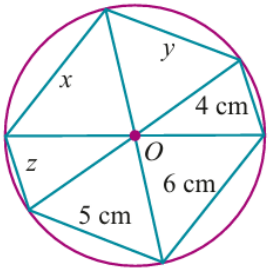
60°

p



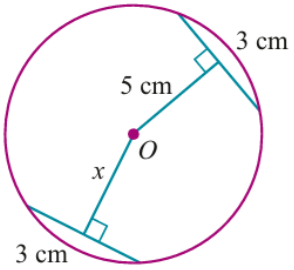
30 cm

q



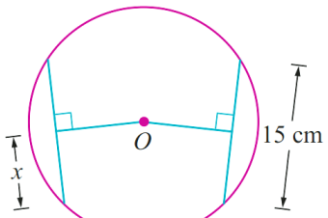
$x = 6$ cm $y = 5$ cm $z = 4$ cm

r



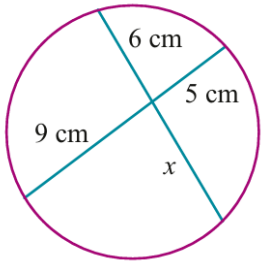
5 cm

s



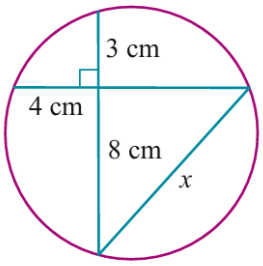
7.5 cm

t



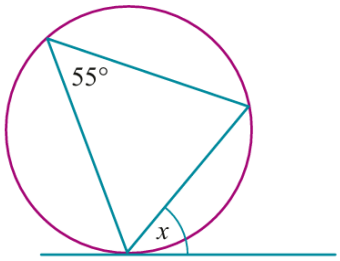
7.5 cm

u



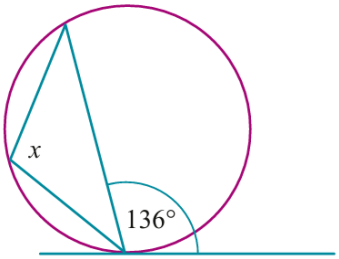
10 cm

v



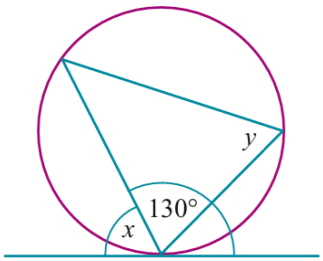
55°

w



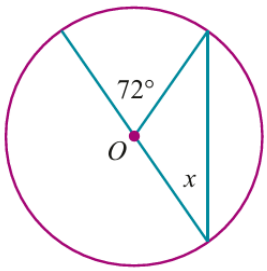
136°

x



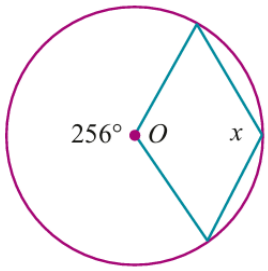
$x = 50°, y = 50°$

y



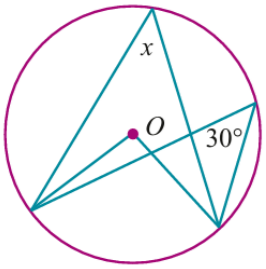
36°

z



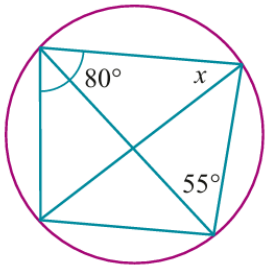
128°

aa



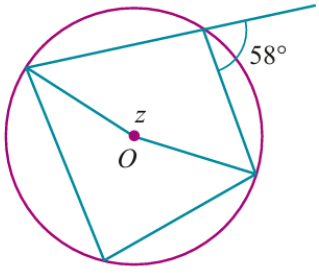
30°

bb



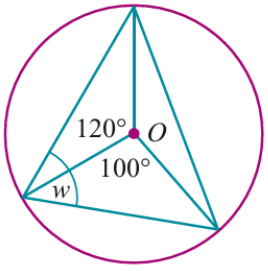
45°

cc



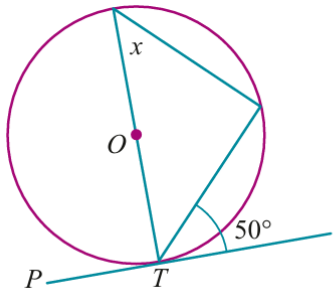
116°

dd



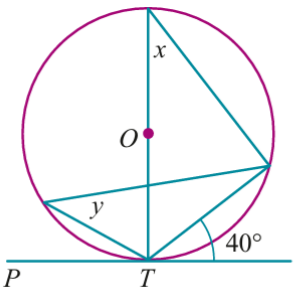
70°

ee



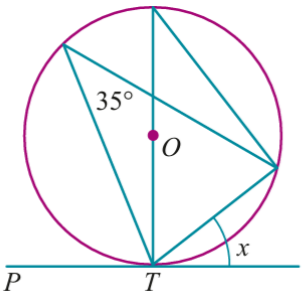
50°

ff



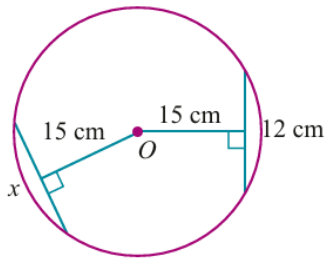
$x = 40°, y = 40°$

gg



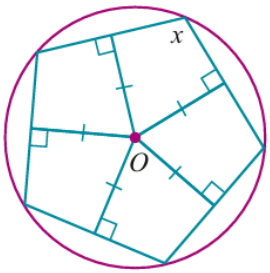
35°

hh



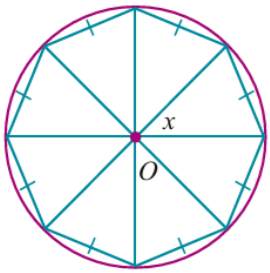
12 cm

ii



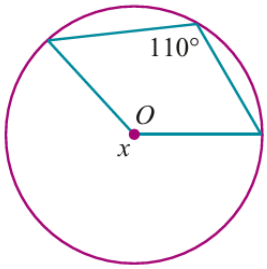
108°

jj



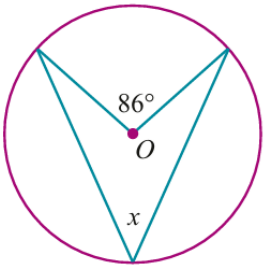
45°

kk



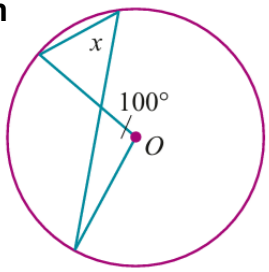
220°

ll



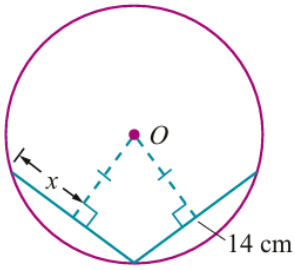
43°

mm



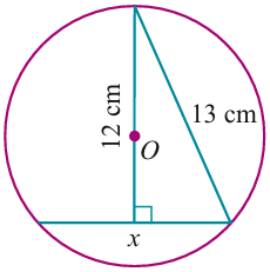
50°

nn



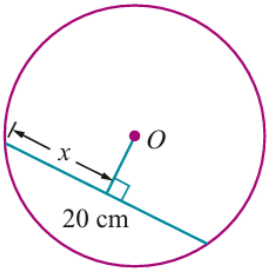
7 cm

oo



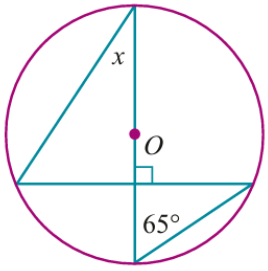
10 cm

pp



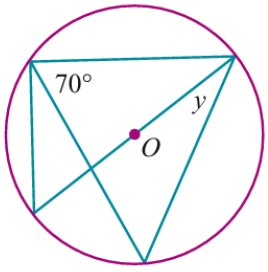
10 cm

qq



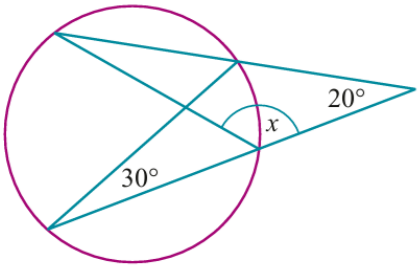
25°

rr



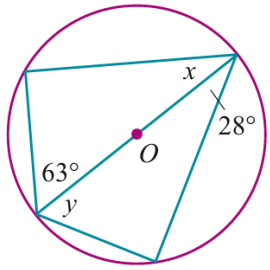
20°

ss



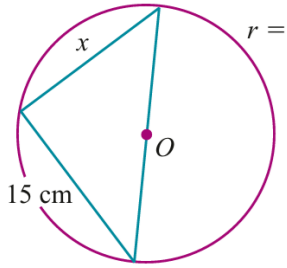
130°

tt



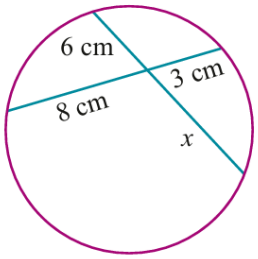
$x = 27°, y = 62°$

uu



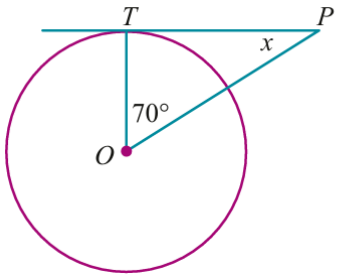
8 cm

vv



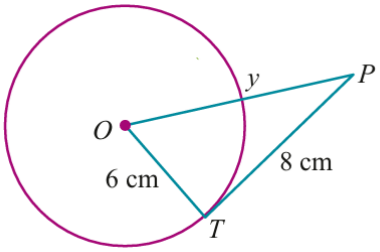
4 cm

ww



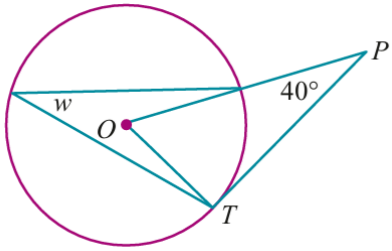
20°

xx



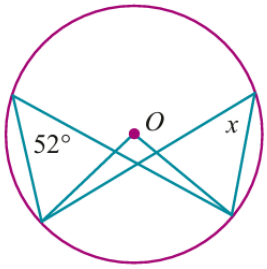
10 cm

yy



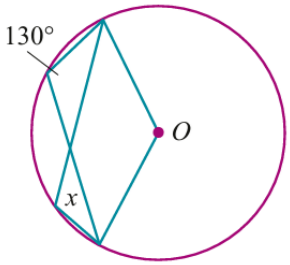
25°

zz



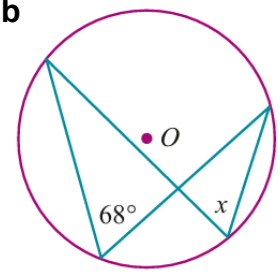
52°

aaa



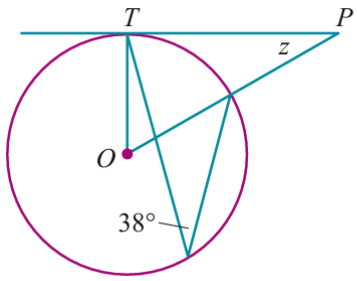
130°

bbb



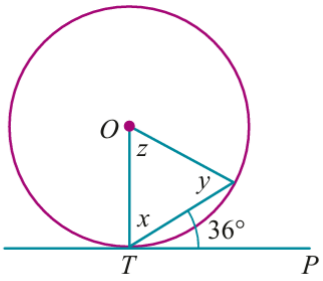
68°

ccc



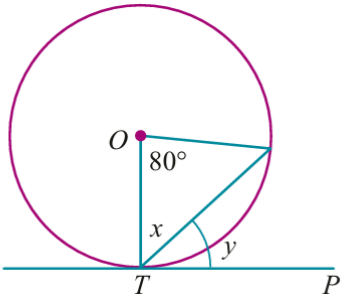
14°

ddd



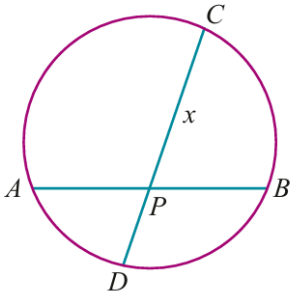
$x = 54°, y = 54°, z = 72°$

eee



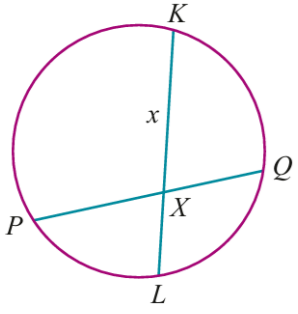
$x = 50°, y = 40°$

fff $AB = 40$ cm, $PD = 25$ cm
CD bisects AB



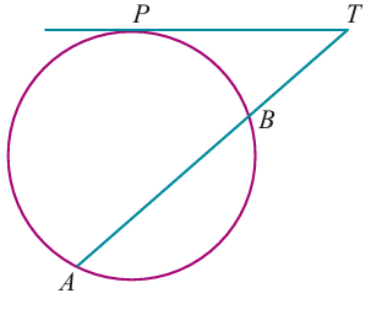
16 cm

ggg $PQ = 24$ cm, $XQ = 8$ cm
$XL = 10$ cm



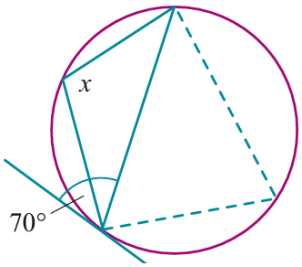
12.8 cm

hhh $TA = 8$ cm, $TB = 2$ cm
Find TP



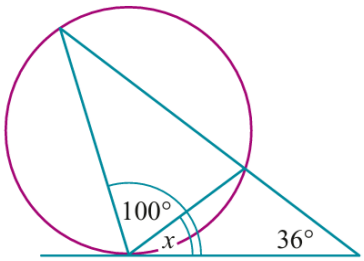
4 cm

iii



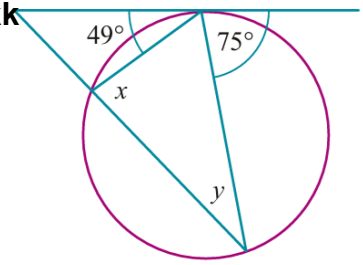
110°

jjj



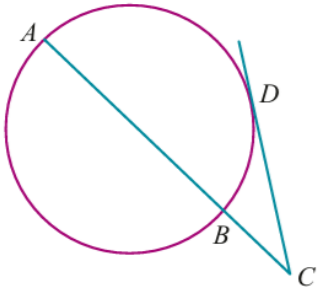
44°

kkk



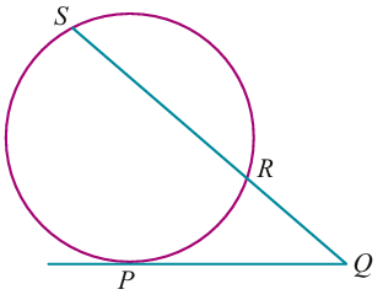
$x = 75°, y = 49°$

lll $CA = 12$ cm, $CB = 3$ cm
Find CD



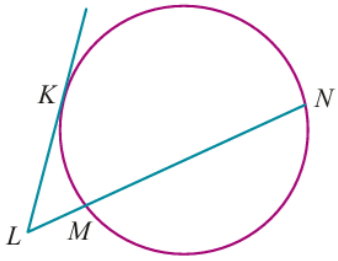
6 cm

mmm $QP = 10$ cm, $QR = 5$ cm
Find QS and RS



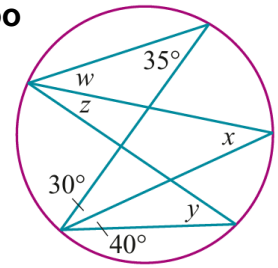
QS = 20 cm, RS = 15 cm

nnn $LK = 12$ cm, $LN = 16$ cm
Find LM and MN



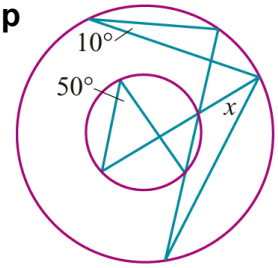
LM = 9 cm, MN = 7 cm

ooo



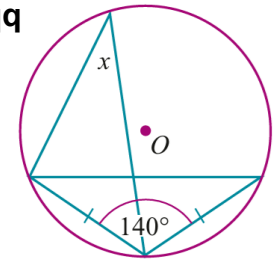
$w = 30°,\ x = 35°, y = 35°, z = 40°$

ppp



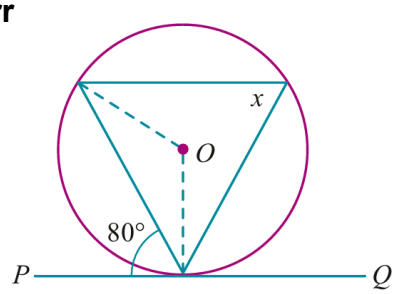
40°

qqq



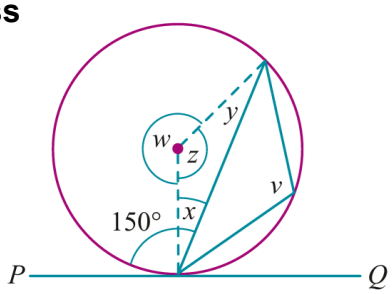
20°

rrr



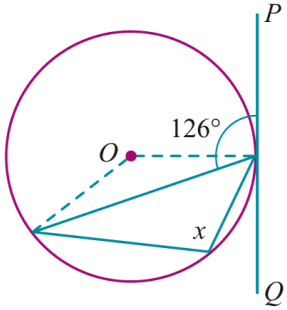
80°

sss



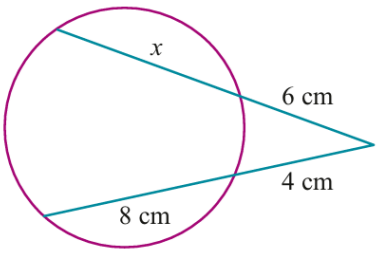
$x = 60°, y = 60°, z = 60°,$
$w = 300°, v = 150°$

ttt



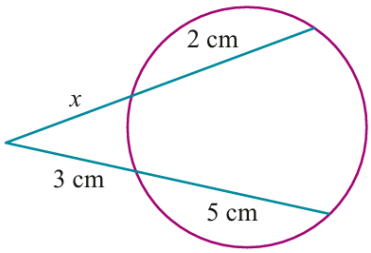
126°

uuu



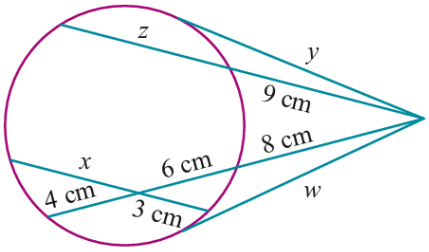
2 cm

vvv



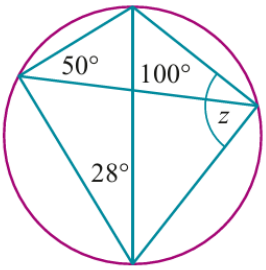
4 cm

www



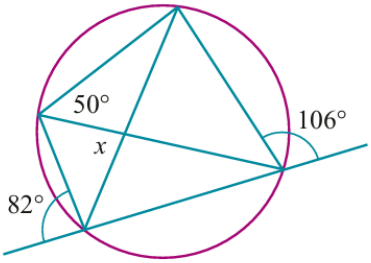
$w = 12$ cm, $x = 8$ cm
$y = 12$ cm, $z = 7$ cm

xxx



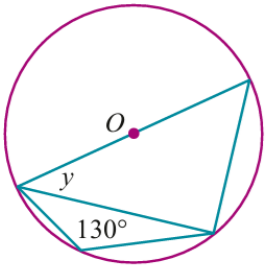
78°

yyy



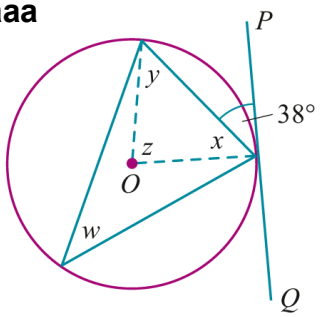
76°

zzz



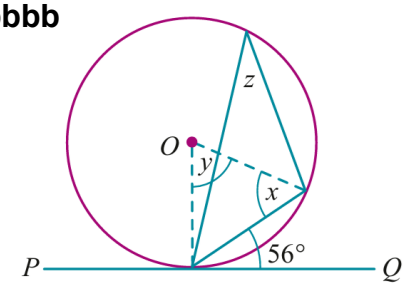
40°

aaaa



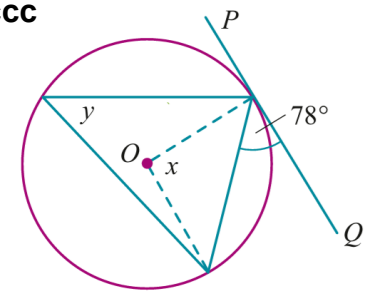
$x = 52°, y = 52°, z = 76°, w = 38°$

bbbb



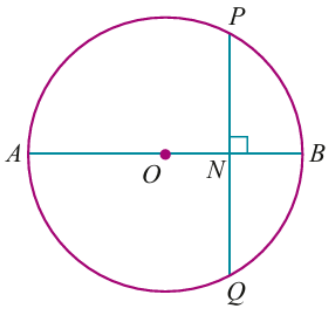
$x = 34°, y = 112°, z = 56°$

cccc



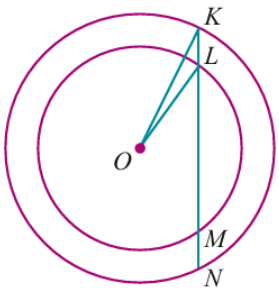
$x = 156°, y = 78°$

2 AN = 18 cm and NB = 8 cm. Find the length of PQ.



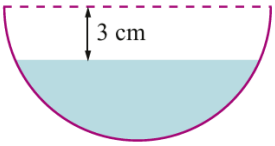
24 cm

3 The figure shows two concentric circles, centre O. OL = 13 cm, OK = 15 cm and LM = 10 cm. Find KN.



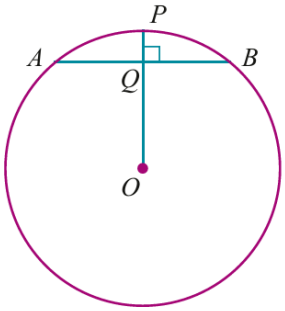
18 cm

4 A hemispherical bowl has a diameter of 10 cm.
Water is poured into the bowl until its surface is 3 cm below the top of the bowl as shown.
Find the diameter of the water surface.



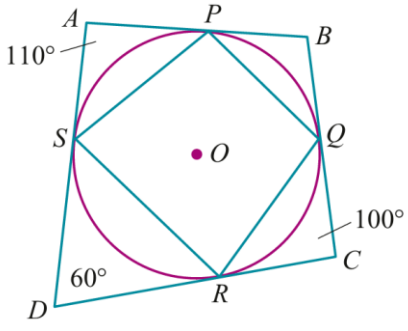
8 cm

5 OP is perpendicular to the chord AB. If AB = 14 cm and PQ = 1 cm, find the radius of the circle.



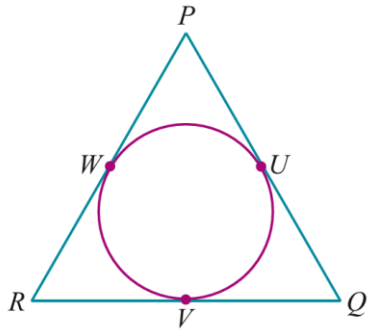
25 cm

6 The circle touches the sides of the quadrilateral ABCD at P, Q, R and S as shown. Find the angles of the quadrilateral PQRS.



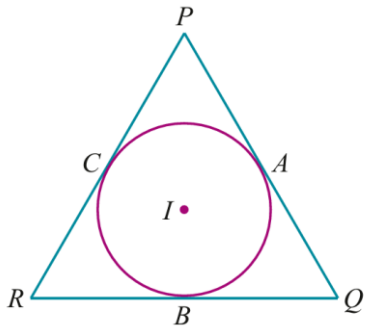
$\angle P = 100°, \angle Q = 95°, \angle R = 80°, \angle S = 85°$

7 PQ, QR and RP are tangents. If PR = 9 cm, RW = 5 cm and RQ = 12 cm, find PQ.



$PQ = 11$ cm

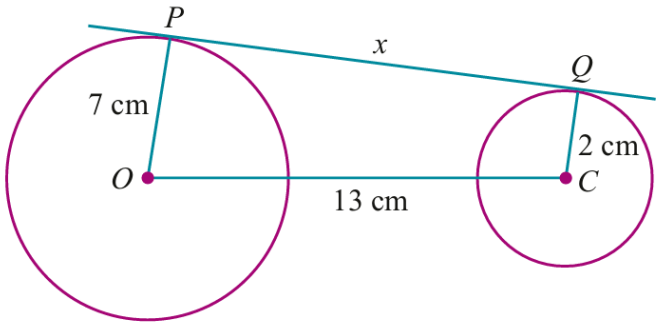
8 I is the incentre of △PQR. RQ = 8 cm, PR = 9 cm and PQ = 7 cm. Find RB.
Hint: set up a system of 3 simultaneous equations.



$RB = 5$ cm

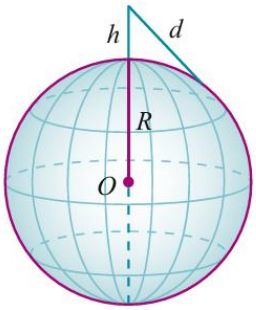
9 Two circles of radii 2 cm and 7 cm have their centres 13 cm apart.

Find the length of the direct common tangent PQ.



12 cm

10 A person of height h metres stands on the surface of Earth, which is assumed to be a sphere of radius R.



a Show that the distance, d, to the horizon is given by $d = \sqrt{h^2 + 2hR}$

$d^2 = h(h + 2R)$

The square of a tangent to a circle from an external point is the product of the intercepts of any secants from the point.

b Calculate the distance across level ground to the horizon that can be seen by a person of height 1.8 m. Use $R = 6400$ km.

4800.000338 m

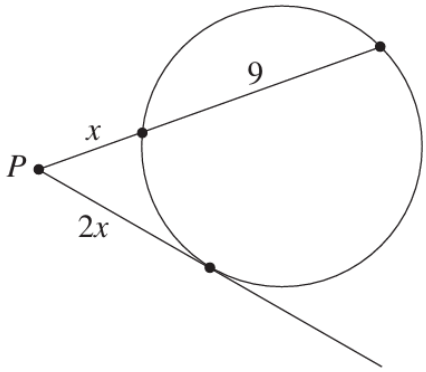
c The formula in part **a** is often given as $d \approx \sqrt{2hr}$. Why might this be the case?

h^2 is extremely small compared to $2hR$.

Exam Questions

1 2019 HSC Extension 1 Band 3

From the point P outside a circle, a secant and a tangent to the circle are constructed a shown in the diagram:



Which equation is satisfied by x?

A. $4x^2 = 9x$

B. $4x^2 = 9 + x$

C. $4x^2 = 9(9 + x)$

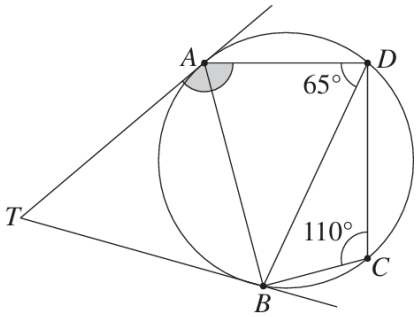
D. $4x^2 = x(9 + x)$

D

2 2017 HSC Extension 1 Band 3

The points A, B, C and D lie on a circle and the tangents at A and B meet at T.

What is the value of $\angle TAD$?



A. 130°

B. 135°

C. 155°

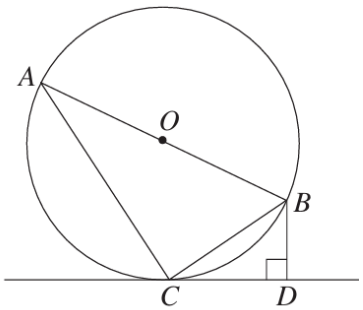
D. 175°

B

3 2019 HSC Extension 1 Band 4

The line segment AB is a diameter of the circle, centred at O. The line CD is the tangent to the circle at C and BD is perpendicular to CD, as shown in the diagram.

Prove that ∠ABC = ∠CBD



Let ∠BCD = α

∴ ∠CAB = α (angle in alt segment)

∠ACB = 90° (angle in a semi-circle)

∴ ∠ABC = 90 – α (angle sum △)

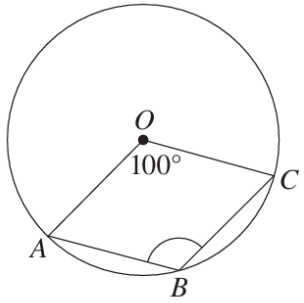
∠CBD = 90 – α (angle sum △)

∴ ∠ABC = ∠CBD

4 2017 HSC Extension 1 Band 4

The points A, B and C lie on a circle with centre O, as shown in the diagram.

The size of $\angle AOC$ is 100°. Find the size of $\angle ABC$, giving reasons.



Reflex $\angle AOC = 360° - 100°$ (angle of revolution)

$= 260°$

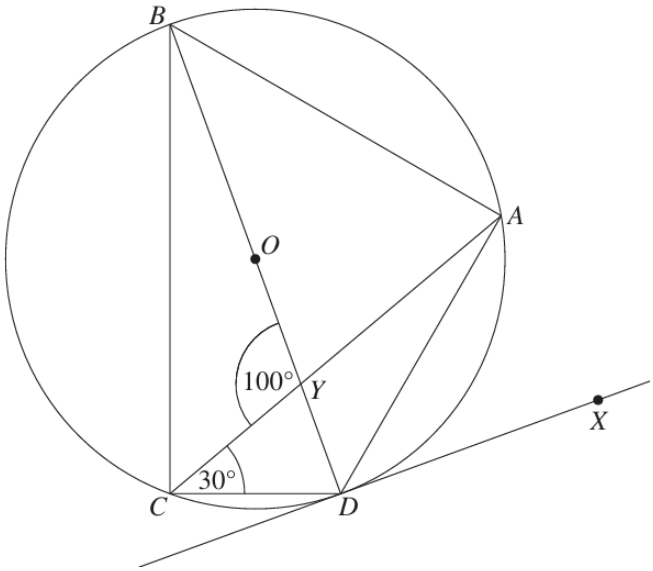
$\angle ABC = \frac{1}{2} \times$ reflex $\angle AOC$ (angle at centre equals twice the angle at circumference standing in the same arc)

$= \frac{260}{2}$

$= 130°$

5 2015 HSC Extension 1 Band 4

aaaIn the diagram, the points A, B, C and D are on the circumference of a circle, whose centre O lies on BD. The chord AC intersects the diameter BD at Y. The tangent at D passes through the point X. It is given that $\angle CYB = 100°$ and $\angle DCY = 30°$



a What is the size of ∠ACB?

∠ACB + ∠ACD = 90° (Angle in a semi-circle is a right angle)

∠ACB + 30° = 90°

∴ ∠ACB = 60°

b What is the size of ∠ADX?

∠ADX = ∠ACD (alternate segment theorem) = 30°

c Find, giving reasons, the size of ∠CAB.

∠CDY + ∠YCD = 100° (∠BYC, exterior ∠ of △CDY)

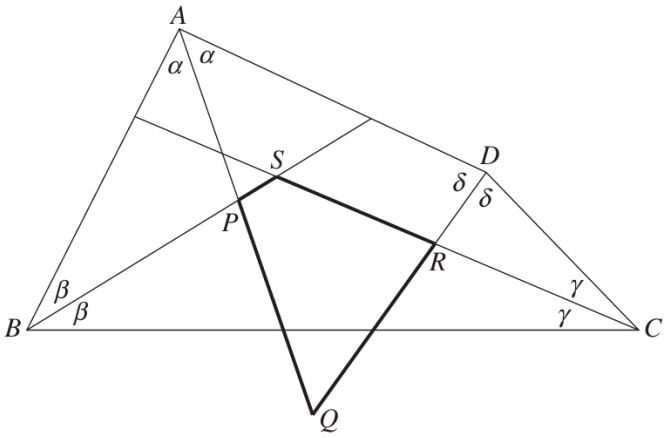
∴ ∠CDY = 70° ∠CAB = ∠CDB (Angles standing on same chord at the circumference are equal)

∴ ∠CAB = 70°

6 2018 HSC Extension 1 Band 5

The diagram shows quadrilateral ABCD and the bisectors of the angles at A, B, C and D. The bisectors at A and B intersect at the point P. The bisectors at A and D meet at Q. The bisectors at C and D meet at R. The bisectors at B and C meet at S.

Show that PQRS is a cyclic quadrilateral.



Let $\alpha = \frac{1}{2}\angle A, \beta = \frac{1}{2}\angle B$

$\gamma = \frac{1}{2}\angle C, \delta = \frac{1}{2}\angle D$

Then $2\alpha + 2\beta + 2\gamma + 2\delta = 360^\circ$ ($\angle$ sum quadrilateral)

$\therefore \alpha + \beta + \gamma + \delta = 180^\circ$

$\angle PSR = \angle BSC = 180^\circ - (\beta + \gamma)$ ($\angle$ sum of $\triangle$BSC)

$\angle PQR = \angle AQD = 180^\circ - (\alpha + \delta)$ ($\angle$ sum of $\triangle$AQD)

$\therefore \angle PSR + \angle PQR = 2 \times 180^\circ - (\alpha + \beta + \gamma + \delta)$

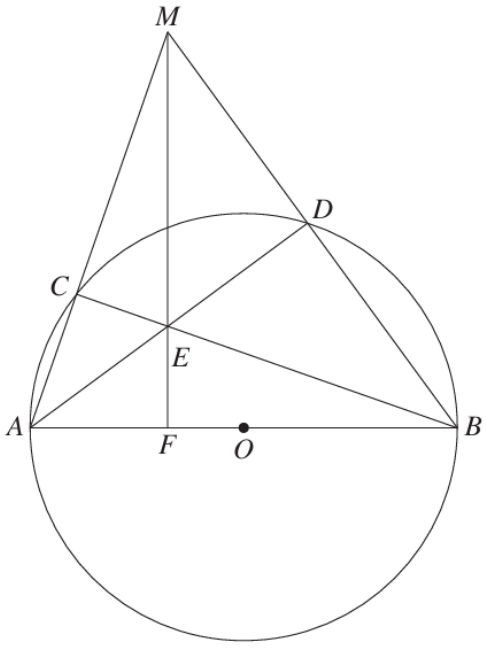
$= 2 \times 180° - 180°$

$= 180°$

$\therefore$ PQRS is cyclic (opposite angles are supplementary)

7 2016 HSC Extension 1 Band 5

The circle centred at O has a diameter AB. From the point M outside the circle the line segments MA and MB are drawn meeting the circle at C and D respectively.
The chords AD and BC meet at E. The line segment ME produced meets the diameter AB at F.



a Show that $CMDE$ is a cyclic quadrilateral.

$\angle ADB = 90°, \angle BCA = 90°$ ($\angle$ in a semicircle)

$\angle MDE = 90°$ ($\angle$ sum, straight line)

$\therefore CMDE$ is a cyclic quadrilateral (Exterior $\angle$ACB equal to opposite interior $\angle$MDE)

b Hence, or otherwise, prove that MF is perpendicular to AB.

Construct CD

Let $\angle ABC = x$

$\therefore \angle ADC = x$ ($\angle$s in the same segment)

Similarly $\angle CME = x$

$\angle CAB = 90^\circ - x$ ($\angle$ sum $\triangle$ABC)

$\therefore \angle MFA = 180^\circ - \angle CME - \angle CAB$ ($\angle$ sum $\triangle$MFA)

$= 180° - x - (90° - x°)$

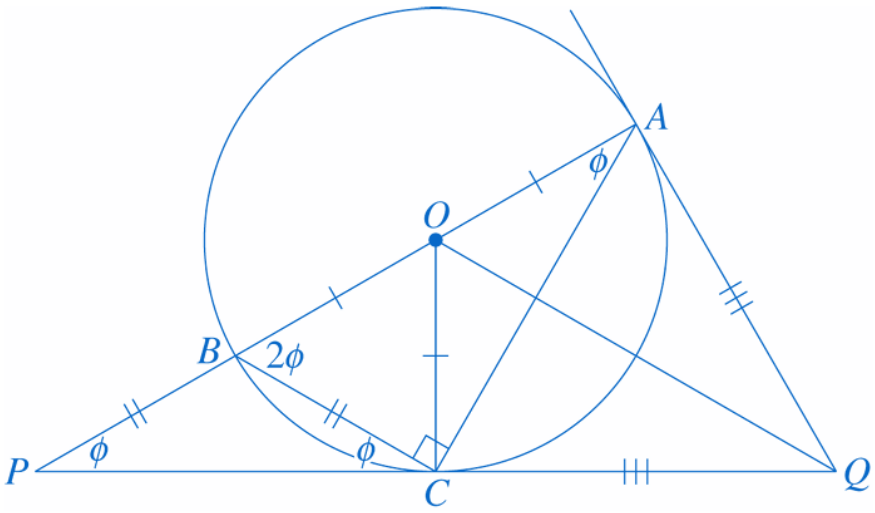
$= 90°$

$\therefore MF \perp AB$

8 2016 HSC Extension 2 Band 5

The circle centred at O has diameter AB. A point P on AB produced is chosen so that PC is a tangent to the circle at C and BP = BC. The tangents to the circle at A and C meet at Q.

Prove that $OP = OQ$.



Construct OC, AC, OQ

Let $\angle BPC$ have size ϕ
$\triangle BPC$ is isosceles so $\angle BCP$ is ϕ
$\angle OBC$ is an external angle
So $\angle OBC = 2\phi$

$\angle BAC$ is angle on chord BC
So $\angle BAC = \phi$ (Angle in alt. seg. and angle between chord and tangent)
But $\triangle BAC$ has a right angle at C (Angle in semicircle)

So $\phi + 2\phi = 90°$

$\phi = 30°$

$OA \perp AQ$ (Angle between tangent and radius)
So $\angle CAQ = 60°$

$QA = QC$ (Tangents from point equal)
So $\angle QCA = 60°$ ($\triangle QAC$ isosceles)

Hence $\angle AQC = 60°$ (Angle sum of $\triangle$)

$\triangle OAQ$ and $\triangle OCQ$ are cong. (sss, $OA = OC$ radii, $AQ = CQ$ above, OQ common)
So $\angle OQC = \frac{60°}{2} = 30° = \angle OPC$

Thus, $\triangle OPQ$ is isosceles
And, $OP = OQ$

Learning Intention

Review (Questions as relevant to the planned examples as possible)

- ☐ Recall
- ☐ Identify

1

	c		**d**
a		**b**	**c**
a	**b**	**c**	**d**

Definition | Formula | Procedure

Identify Language | Decision

Scenario



Testing Sequence

a			**b**	
a		**b**		**c**
a	**b**		**c**	**d**

Interpret Transformation

Scenario

Testing Sequence

a			
b			**c**
d		**e**	**f**
d	**e**	**f**	**g**

Example Success Criteria

Scenario

Worked example	Checks for listening

Guided Practice

Test the same skill as the I Do but still require genuine thinking - not just pattern matching.

Sequence of examples varying critical features

a

b		**c**
d		**e**
g	**h**	**i**

Key Ideas

1. Fact atoms
2. Procedure

$\text{proj}_{\boldsymbol{b}}\boldsymbol{a}$

$\text{proj}_{\boldsymbol{u}}\boldsymbol{v}$

FOUNDATION

e

Type equation here.

f

Type equation here.

i

ii

dddd

Type equation here.

eeee

Type equation here.

ffff

Type equation here.

i

ii

iii

a

b

c

d

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Desmos

GeoGebra

Symbols:

✓ —

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